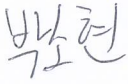


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☐ Controlled
☐ Uncontrolled

	Prepared	Reviewed	Approved
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Position	RA Specialist	-	QMR
Date	2026. 01. 12	-	2026. 01. 12
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1. INTRODUCTION

This Common Submission Dossier Template (CSDT) has been prepared to support the regulatory submission for the medical device CUVIS-spine, manufactured by CUREXO, INC., for market authorization in Vietnam.

This dossier is compiled in accordance with the principles of the ASEAN Medical Device Directive (AMDD) and the applicable national regulatory requirements of Vietnam, including Decree No. 98/2021/ND-CP and its related implementing regulations under the authority of the Ministry of Health (MOH).

The objective of this CSDT is to provide comprehensive and objective evidence demonstrating that the subject medical device complies with the applicable Essential Principles of Safety and Performance, and that it is safe, effective, and performs as intended for its specified use.

This dossier contains detailed information on the device description and specification, intended use, design and manufacturing processes, risk management, product verification and validation, clinical evidence (where applicable), labeling, and post-market surveillance considerations. The content and structure of this submission follow the CSDT format to facilitate efficient and consistent review by the Vietnamese regulatory authority.

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2. REFERENCES

The following regulations, standards, and guidance documents were referenced in the preparation of this Common Submission Dossier Template (CSDT) for the medical device CUVIS-spine, to support demonstration of compliance with the applicable regulatory requirements and the Essential Principles of Safety and Performance in Vietnam.

2.1. ASEAN and International Regulations and Guidance

- ASEAN Medical Device Directive (AMDD).
- ASEAN Common Submission Dossier Template (CSDT) for Medical Devices.
- ASEAN Essential Principles of Safety and Performance for Medical Devices.

2.2. Vietnam

- Decree No. 98/2021/ND-CP on the Management of Medical Devices.
- Decree No. 07/2023/ND-CP amending and supplementing Decree No. 98/2021/ND-CP.
- Relevant circulars and guidance documents issued by the Ministry of Health (MOH) of Vietnam.

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3. EXECUTIVE SUMMARY

3.1. Overview

- (1) Medical Device Name : CUVIS-spine
- (2) Model Name : CS200
- (3) Product Type : Pedicle Screw Guide System
- (4) Manufacturer : CUREXO, Inc.
- (5) Description of medical device

CUVIS-spine is a mobile system mainly comprising the robotic arm, the main console and the staff console as an option. The robotic arm is positioned on the floor near the side of the surgical table. The location of the main console or the staff console is appropriately determined considering the user preference and the environments.

CUVIS-spine is a pedicle screw guide system which consists of Robotic Arm, Main Console, Staff Console, Guide bush, Source Calibrator, Registration Tool, Registration Tool Adapter, Drape, Marker Ball, Patient Marker, Clamp, Adapter, Pin, Pin vise, Slide Hammer, Marker Driver, Detector calibrator, Dilator, Serration-tip dilator, Bur, Awl, Probe, Lenke probe, Tapper, Stylet tapper, Screwdriver, Instrument container, Handle.

3.2. Commercial Marketing History and Regulatory History

No.	Country	Device name	Dates of introduction into market
1	Rep. of Korea	CUVIS-spine (Model: CS200)	23 December, 2022
2	US	CUVIS-spine (Model:CS200)	22 June, 2023
3	Indonesia	CUVIS-spine (Model:CS200)	10 May, 2023
4	Taiwan	CUVIS-spine (Model:CS200)	24 December, 2024
5	Malaysia	CUVIS-spine (Model:CS200)	22 April, 2025

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3.3. Intended Uses and Indications in Labelling

CUVIS-spine is intended for use as an aid for precisely locating anatomical structures and for the spatial positioning and orientation of guide bush to be used by surgeons for navigating and/or guiding compatible surgical instruments in open or percutaneous pedicle screw placement provided that the required markers and rigid patient anatomy can be identified on 3D or 2D image. Details may refer to **Appendix 1**.

- ✓ Appendix 1-1. User Manual (ENG) (Doc No. 310023)
- ✓ Appendix 1-2. System Labeling & Packaging Specification (Doc No.310031)

3.4. List of the Regulatory Approval or Marketing Clearance Obtained

No.	Country	Certificate No.	Dates of introduction into market
1	Rep. of Korea	제인 19-4993호	23 December, 2022
2	US	K223558	22 June, 2023
3	Indonesia	KEMENKES RI AKL 21002320133	10 May, 2023
4	Taiwan	037691	24 December, 2024
5	Malaysia	GB3982625-201723	22 April, 2025

An attestation to this effect has been attached as an **Appendix 2**.

- ✓ Appendix 2-1. Certificate of Republic of Korea
- ✓ Appendix 2-2. Certificate of US
- ✓ Appendix 2-3. Certificate of Indonesia
- ✓ Appendix 2-4. Certificate of Taiwan
- ✓ Appendix 2-5. Certificate of Malaysia

3.5. Status of Any Pending Request for Market Clearance

There is no pending request for market clearance.

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3.6. Important Safety and Performance related information

There has been no reportable adverse events and field safety corrective actions (FSCA) to date for the CUVIS-spine since its first introduction into the global market. An attestation to this effect has been attached as an **Appendix 3**.

- ✓ Appendix 3-1. Declaration of AE and FSCA

3.7. Declaration on Labelling, Packaging and IFU

CUREXO hereby declares that the labelling, packaging, and Instructions for Use (IFU) for the medical device CUVIS-spine have been prepared in compliance with the applicable regulatory requirements of Vietnam, including Decree No. 98/2021/ND-CP and relevant regulations issued by the Ministry of Health (MOH).

Samples of the labelling, packaging, and IFU are included in the **Appendix 1**.

- ✓ Appendix 1-1. User Manual (ENG) (Doc No. 310023)
- ✓ Appendix 1-2. System Labeling & Packaging Specification (Doc No.310031)

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4. RELEVANT ESSENTIAL PRINCIPLES AND RULES USED TO DEMONSTRATE CONFORMITY

Clause	MDR 2012, Third Schedule (Appendix 1): Essential Principle of Safety and Performance (EPSP)	Applicable to the medical device?	Rule of Conformity	Identity of Specific Documents
1.0 General Requirements				
1.1	Medical devices should be designed and manufactured in such a way that, when used under the conditions and for the purposes intended and, where applicable, by virtue of the technical knowledge, experience, education or training of intended users, they will not compromise the clinical condition or the safety of patients, or the safety and health of users or, where applicable, other persons, provided that any risks which may be associated with their use constitute acceptable risks when weighed against the benefits to the patient and are compatible with a high level of protection of health and safety.	A	ISO 13485:2016 ISO 14971:2019 IEC 60601-1:2005/A1:2012 /A2:2020 EN 60601-1-2:2015/A1:2021 IEC 60601-1-6:2010+A1:2013 IEC 80601-2-77:2019 IEC TS 60601-4-2:2024	Quality Manual (CX-001, Rev 22, 2025.03.01) Risk Management Report (310005, Rev 9, 2025.09.24) Compliance Test report - EMC Test Report (310403, Rev 3, 2025.08.07) - Safety Test Report (310404, Rev 3, 2024.07.22) Usability Evaluation Report (310036, Rev 1, 2022.04.30) Software Validation Report

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Clause	MDR 2012, Third Schedule (Appendix 1): Essential Principle of Safety and Performance (EPSP)	Applicable to the medical device?	Rule of Conformity	Identity of Specific Documents
			IEC 62366-1:2015 IEC 62304:2006/A1:2015 MEDDEV 2.7.1(Rev.4) MEDDEV 2.12/2 (Rev.2)	(310027, Rev 12, 2025.09.30) Clinical Evaluation Report (300024, Rev 8, 2023.07.04) User Manual (310023, Rev 15, 2025.10.15)
1.2	The solutions adopted by the manufacturer for the design and manufacture of the devices should conform to safety principles, taking account of the generally acknowledged state of the art. When risk reduction is required, the manufacturer should control the risk(s) so that the residual risk(s) associated with each hazard is judged acceptable. The manufacturer should apply the following principles in the priority order listed:	A	ISO 13485:2016 ISO 14971:2019 IEC 62304:2006/A1:2015	Quality Manual (CX-001, Rev 22, 2025.03.01) Risk Management Report (310005, Rev 9, 2025.09.24) Software Validation Report (310027, Rev 12, 2025.09.30)

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Clause	MDR 2012, Third Schedule (Appendix 1): Essential Principle of Safety and Performance (EPSP)	Applicable to the medical device?	Rule of Conformity	Identity of Specific Documents
	a) identify known or foreseeable hazards and estimate the associated risks arising from the intended use and foreseeable misuse; b) eliminate risks as far as reasonably practicable through inherently safe design and manufacture; c) reduce as far as is reasonably practicable the remaining risks by taking adequate protection measures, including alarms; d) inform users of any residual risks.			
1.3	Devices shall achieve the performance intended by the manufacturer and be designed, manufactured and packaged in such a way that they are suitable for one or more of the functions within the scope of the definition of a medical device applicable in each jurisdiction.	A	ISO 13485:2016 IEC 60601-1:2005/A1:2012 /A2:2020 EN 60601-1-2:2015/A1:2021 IEC 60601-1-	Quality Manual (CX-001, Rev 22, 2025.03.01) Compliance Test report - EMC Test Report (310403, Rev 3, 2025.08.07) - Safety Test Report (310404, Rev 3, 2024.07.22)

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Clause	MDR 2012, Third Schedule (Appendix 1): Essential Principle of Safety and Performance (EPSP)	Applicable to the medical device?	Rule of Conformity	Identity of Specific Documents
			6:2010/A1:2013/A2:2020 IEC 80601-2-77:2019 IEC TS 60601-4-2:2024 IEC 62366-1:2015/AMD1:2020 IEC 62304:2006/A1:2015	Usability Evaluation Report (310036, Rev 1, 2022.04.30) Software Validation Report (310027, Rev 12, 2025.09.30)
1.4	The characteristics and performances referred to in Clauses 1.1, 1.2 and 1.3 shall not be adversely affected to such a degree that the health or safety of the patient or the user and, where applicable, of other persons are compromised during the lifetime of the device, as indicated by the manufacturer, when the device is subjected to the stresses which can occur during normal conditions of use and has been properly maintained in accordance with the manufacturer's instructions.	A	ISO 13485:2016 IEC 60601-1:2005/A1:2012 /A2:2020 EN 60601-1-2:2015/A1:2021 IEC 60601-1-	Quality Manual (CX-001, Rev 22, 2025.03.01) Compliance Test report - EMC Test Report (310403, Rev 3, 2025.08.07) - Safety Test Report (310404, Rev 3, 2024.07.22)

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Clause	MDR 2012, Third Schedule (Appendix 1): Essential Principle of Safety and Performance (EPSP)	Applicable to the medical device?	Rule of Conformity	Identity of Specific Documents
			6:2010/A1:2013/A2:2020 IEC 80601-2-77:2019 IEC TS 60601-4-2:2024 IEC 62366-1:2015/AMD1:2020 IEC 62304:2006/A1:2015 ISO 14971:2019	Usability Evaluation Report (310036, Rev 1, 2022.04.30) Software Validation Report (310027, Rev 12, 2025.09.30) Risk Management Report (310005, Rev 9, 2025.09.24)
1.5	The devices shall be designed, manufactured and packed in such a way that their characteristics and performances during their intended use will not be adversely affected under transport and storage conditions (for example, fluctuations of temperature and humidity) taking account of the instructions and information provided by the manufacturer.	A	ISO 13485:2016 ISO 14971:2019 EN 1041:2008 ISO 15223-1:2021	Quality Manual (CX-001, Rev 22, 2025.03.01) Risk Management Report (310005, Rev 9, 2025.09.24) Labeling & Packaging

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Clause	MDR 2012, Third Schedule (Appendix 1): Essential Principle of Safety and Performance (EPSP)	Applicable to the medical device?	Rule of Conformity	Identity of Specific Documents
				Specification (310031, Rev 11, 2025.09.29) User Manual (310023, Rev 15, 2025.10.15)
1.6	The benefits must be determined to outweigh any undesirable side effects for the performances intended.	A	ISO 14971:2019 MEDDEV 2.7.1 Rev.4	Risk Management Report (310005, Rev 9, 2025.09.24) Clinical Evaluation Report (300024, Rev 8, 2023.07.04)
Design and manufacturing principles				
2.1 Chemical, physical and biological properties				
2.1.1	The devices should be designed and manufactured in such a way as to ensure the characteristics and performance referred to in Clauses 1.1 to 1.6 of the 'General Principles'. Particular attention should be paid to:	A	ISO 13485:2016 IEC 60601- 1:2005/A1:2012 /A2:2020	Quality Manual (CX-001, Rev 22, 2025.03.01) Compliance Test report - EMC Test Report

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Clause	MDR 2012, Third Schedule (Appendix 1): Essential Principle of Safety and Performance (EPSP)	Applicable to the medical device?	Rule of Conformity	Identity of Specific Documents
	a) the choice of materials used, particularly as regards toxicity and, where appropriate, flammability; b) the compatibility between the materials used and biological tissues, cells, body fluids, and specimens, taking account of the intended purpose of the device; c) the choice of materials used should reflect, where appropriate, matters such as hardness, wear and fatigue strength.		EN 60601-1-2:2015/A1:2021 IEC 60601-1-6:2010/A1:2013/A2:2020 IEC 80601-2-77:2019 IEC TS 60601-4-2:2024 ISO 14971:2019 ISO 10993-1:2018 ISO 10993-5:2009 ISO 10993-10:2010 ISO 10993-11:2006	(310403, Rev 3, 2025.08.07) - Safety Test Report (310404, Rev 3, 2024.07.22) Risk Management Report (310005, Rev 9, 2025.09.24) Biocompatibility Evaluation Report (300059, Rev 8, 2025.05.09)

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Clause	MDR 2012, Third Schedule (Appendix 1): Essential Principle of Safety and Performance (EPSP)	Applicable to the medical device?	Rule of Conformity	Identity of Specific Documents
2.1.2	The devices should be designed, manufactured and packed in such a way as to minimize the risk posed by contaminants and residues to the persons involved in the transport, storage and use of the devices and to patients, taking account of the intended purpose of the product. Particular attention should be paid to tissues exposed and to the duration and frequency of exposure.	A	ISO 13485:2016 IEC 60601-1:2005/A1:2012 /A2:2020 EN 60601-1-2:2015/A1:2021 IEC 60601-1-6:2010/A1:2013/A2:2020 IEC 80601-2-77:2019 IEC TS 60601-4-2:2024 ISO 14971:2019 ISO 10993-1:2018	Quality Manual (CX-001, Rev 22, 2025.03.01) Compliance Test report - EMC Test Report (310403, Rev 3, 2025.08.07) - Safety Test Report (310404, Rev 3, 2024.07.22) Risk Management Report (310005, Rev 9, 2025.09.24) Biocompatibility Evaluation Report (300059, Rev 8, 2025.05.09)

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Clause	MDR 2012, Third Schedule (Appendix 1): Essential Principle of Safety and Performance (EPSP)	Applicable to the medical device?	Rule of Conformity	Identity of Specific Documents
			ISO 10993-10:2010 ISO 10993-11:2006	
2.1.3	The devices should be designed and manufactured in such a way that they can be used safely with the materials, substances and gases with which they enter into contact during their normal use or during routine procedures; if the devices are intended to administer medicinal products they should be designed and manufactured in such a way as to be compatible with the medicinal products concerned according to the provisions and restrictions governing these products and that their performance is maintained in accordance with the intended use.	NA (The device does not contact any substances, gases and medicinal products.)		
2.1.4	Where a device incorporates, as an integral part, a substance which, if used separately, may be considered to be a medicinal product/drug as defined in the relevant legislation that applies within that jurisdiction and which is liable to act	NA (The device does not incorporate a substance		

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Clause	MDR 2012, Third Schedule (Appendix 1): Essential Principle of Safety and Performance (EPSP)	Applicable to the medical device?	Rule of Conformity	Identity of Specific Documents
	upon the body with action ancillary to that of the device, the safety, quality and usefulness of the substance should be verified, taking account of the intended purpose of the device.	which be considered to be medicinal product.)		
2.1.5	The devices should be designed and manufactured in such a way as to reduce as far as reasonably practicable and appropriate the risks posed by substances that may leach or leak from the device.	NA (The device does not incorporate a substance which be considered to be medicinal product.)		
2.1.6	Devices should be designed and manufactured in such a way as to reduce as far as reasonably practicable and appropriate risks posed by the unintentional ingress or egress of substances into or from the device taking into account the device and the nature of the environment in which it is intended to be used.	A	ISO 14971:2019	Risk Management Report (310005, Rev 9, 2025.09.24)

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Clause	MDR 2012, Third Schedule (Appendix 1): Essential Principle of Safety and Performance (EPSP)	Applicable to the medical device?	Rule of Conformity	Identity of Specific Documents
2.2 Infection and microbial contamination				
2.2.1	The devices and manufacturing processes should be designed in such a way as to eliminate or to reduce as far as reasonably practicable and appropriate the risk of infection to patients, users and, where applicable, other persons. The design should: a) allow easy handling; b) and, where necessary: c) reduce as far as reasonably practicable and appropriate any microbial leakage from the device and/or microbial exposure during use; d) prevent microbial contamination of the device, or specimen where applicable, by the patient, user or other person.	A (The device is used together with the disposable sterilized parts and reusable parts through user sterilization.)	ISO 13485:2016 ISO 14971:2019	Quality Manual (CX-001, Rev 22, 2025.03.01) Risk Management Report (310005, Rev 9, 2025.09.24)
2.2.2	Where a device incorporates substances of biological origin, the risk of infection must be reduced as far as reasonably practicable and appropriate by selecting appropriate sources, donors and substances and by using, as appropriate,	NA (The device does not incorporate any substances		

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Clause	MDR 2012, Third Schedule (Appendix 1): Essential Principle of Safety and Performance (EPSP)	Applicable to the medical device?	Rule of Conformity	Identity of Specific Documents
	validated inactivation, conservation, test and control procedures.	of biological origin.)		
2.2.3	In some jurisdictions products incorporating tissues, cells and substances of non-human origin may be considered medical devices. In this case, such tissues, cells and substances should originate from animals that have been subjected to veterinary controls and surveillance adapted to the intended use of the tissues. National regulations may require that the manufacturer and/or the Regulatory Authority retain information on the geographical origin of the animals. Processing, preservation, testing and handling of tissues, cells and substances of animal origin should be carried out so as to provide optimal safety. In particular, safety with regard to viruses and other transmissible agents should be addressed by implementation of validated	NA (The device does not incorporate any tissues, cells and substances of non-human origin.)		

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Clause	MDR 2012, Third Schedule (Appendix 1): Essential Principle of Safety and Performance (EPSP)	Applicable to the medical device?	Rule of Conformity	Identity of Specific Documents
	methods of elimination or inactivation in the course of the manufacturing process.			
2.2.4	In some jurisdictions products incorporating human tissues, cells and substances may be considered medical devices. In this case, the selection of sources, donors and/or substances of human origin, the processing, preservation, testing and handling of tissues, cells and substances of such origin should be carried out so as to provide optimal safety. In particular, safety with regard to viruses and other transmissible agents should be addressed by implementation of validated methods of elimination or inactivation in the course of the manufacturing process.	NA (The device does not incorporate any tissues, cells and substances of human origin.)		
2.2.5	Devices labelled as having a special microbiological state should be designed, manufactured and packed to ensure they remain so when placed on the market and remain so under the	NA (The device does not have any special		

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Clause	MDR 2012, Third Schedule (Appendix 1): Essential Principle of Safety and Performance (EPSP)	Applicable to the medical device?	Rule of Conformity	Identity of Specific Documents
	transport and storage conditions specified by the manufacturer.	microbiological state)		
2.2.6	Devices delivered in a sterile state should be designed, manufactured and packed in a non-reusable pack, and/or according to appropriate procedures, to ensure that they are sterile when placed on the market and remain sterile, under the transport and storage conditions indicated by the manufacturer, until the protective packaging is damaged or opened.	A (This system is not a sterile medical device. But, it is used together with the disposable sterilized parts and reusable parts through user sterilization.)	ISO 17665-1:2006 ISO 11135:2014	User cleaning & sterilization evaluation report (300051, Rev 9, 2025.05.09) Ethylene Oxide Sterilization Validation Report (300075, Rev 1, 2021.08.30)
2.2.7	Devices labelled either as sterile or as having a special microbiological state should have been processed, manufactured and, if applicable, sterilized by appropriate, validated methods.	A (This system is not a sterile medical device. But, it is used	ISO 17665-1:2006 ISO 11135:2014	User cleaning & sterilization evaluation report (300051, Rev 9, 2025.05.09) Ethylene Oxide Sterilization

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		together with the disposable sterilized parts and reusable parts through user sterilization.)		Validation Report (300075, Rev 1, 2021.08.30)
2.2.8	Devices intended to be sterilized should be manufactured in appropriately controlled (e.g. environmental) conditions.	A (This system is not a sterile medical device. But, it is used together with the disposable sterilized parts and reusable parts through user sterilization.)	ISO 17665-1:2006 ISO 11135:2014	User cleaning & sterilization evaluation report (300051, Rev 9, 2025.05.09) Ethylene Oxide Sterilization Validation Report (300075, Rev 1, 2021.08.30)

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2.2.9	Packaging systems for non-sterile devices should keep the product without deterioration at the level of cleanliness stipulated and, if the devices are to be sterilized prior to use, minimize the risk of microbial contamination; the packaging system should be suitable taking account of the method of sterilization indicated by the manufacturer.	A (This system is not a sterile medical device. But, it is used together with the disposable sterilized parts and reusable parts through user sterilization.)	ISO 17665-1:2006 ISO 11135:2014	User cleaning & sterilization evaluation report (300051, Rev 9, 2025.05.09) Ethylene Oxide Sterilization Validation Report (300075, Rev 1, 2021.08.30)
2.2.10	The packaging and/or label of the device should distinguish between identical or similar product	NA (There are no identical or similar products sold in both sterile and non-sterile condition.in this		

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		system)		
2.3 Manufacturing and environmental properties				
2.3.1	If the device is intended for use in combination with other devices or equipment, the whole combination, including the connection system should be safe and should not impair the specified performance of the devices. Any restrictions on use applying to such combinations should be indicated on the label and/or in the instructions for use.	A	ISO 14971:2019 ISO 15223-1:2021	Performance test report (300065, Rev 2, 2024.09.23) Risk Management Report (310005, Rev 9, 2025.09.24) User Manual (310023, Rev 15, 2025.10.15)
2.3.2	Devices should be designed and manufactured in such a way as to remove or reduce as far as reasonably practicable and appropriate: a) the risk of injury, in connection with their physical features, including the volume/pressure ratio, dimensional and where appropriate ergonomic features; b) risks connected with reasonably foreseeable external influences or environmental	A	ISO 13485:2016 ISO 14971:2019 IEC 62366-1:2015/AMD1:2020 IEC TR 62366-2:2016	Quality Manual (CX-001, Rev 22, 2025.03.01) Risk Management Report (310005, Rev 9, 2025.09.24) Usability Evaluation Report (310036, Rev 1, 2022.04.30)

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	conditions, such as magnetic fields, external electrical and electromagnetic effects, electrostatic discharge, pressure, humidity, temperature or variations in pressure and accelerations; c) the risks connected to their use in conjunction with materials, substances and gases with which they may come into contact during normal conditions of use; d) the risks of accidental penetration of substances into the device; e) the risk of incorrect identification of specimens; f) the risks of reciprocal interference with other devices normally used in the investigations or for the treatment given; g) risks arising where maintenance or calibration are not possible (as with implants), from ageing of materials used or loss of accuracy of any measuring or control mechanism.		IEC 60601-1:2005/A1:2012 /A2:2020 EN 60601-1-2:2015/A1:2021 IEC 60601-1-6:2010/A1:2013/A2:2020 IEC 80601-2-77:2019 IEC TS 60601-4-2:2024	Compliance Test report - EMC Test Report (310403, Rev 3, 2025.08.07) - Safety Test Report (310404, Rev 3, 2024.07.22)

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2.3.3	Devices should be designed and manufactured in such a way as to minimize the risks of fire or explosion during normal use and in single fault condition. Particular attention should be paid to devices whose intended use includes exposure to or use in association with flammable substances or substances which could cause combustion.	A	ISO 13485:2016 ISO 14971:2019 IEC 60601-1:2005/A1:2012 /A2:2020 EN 60601-1-2:2015/A1:2021 IEC 60601-1-6:2010/A1:2013/A2:2020 IEC 80601-2-77:2019 IEC TS 60601-4-2:2024 EN 1041:2008	Quality Manual (CX-001, Rev 22, 2025.03.01) Risk Management Report (310005, Rev 9, 2025.09.24) Compliance Test report - EMC Test Report (310403, Rev 3, 2025.08.07) - Safety Test Report (310404, Rev 3, 2024.07.22) User Manual (310023, Rev 15, 2025.10.15)

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2.3.4	Devices must be designed and manufactured in such a way as to facilitate the safe disposal of any waste substances.	A	EN 1041:2008 ISO 15223-1:2021	Labeling & Packaging Specification (310031, Rev 11, 2025.09.29) User Manual (310023, Rev 15, 2025.10.15)
2.4 Devices with a diagnostic or measuring function				
2.4.1	Devices with a measuring function, where inaccuracy could have a significant adverse effect on the patient, should be designed and manufactured in such a way as to provide sufficient accuracy, precision and stability for their intended purpose of the device. The limits of accuracy should be indicated by the manufacturer.	NA (The device does not have any diagnostic or measuring function.)		
2.4.2	Diagnostic devices should be designed and manufactured in such a way as to provide sufficient accuracy, precision and stability for their intended	NA (The device does not have		

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	use, based on appropriate scientific and technical methods. In particular the design should address sensitivity, specificity, trueness, repeatability, reproducibility, control of known relevant interference and limits of detection, as appropriate.	any diagnostic or measuring function.)		
2.4.3	Where the performance of devices depends on the use of calibrators and/or control materials, the traceability of values assigned to such calibrators and/or control materials should be assured through a quality management system.	NA (The device does not have any diagnostic or measuring function.)		
2.4.4	Any measurement, monitoring or display scale should be designed in line with ergonomic principles, taking account of the intended purpose of the device.	NA (The device does not have any diagnostic or measuring function.)		

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2.4.5	Wherever possible values expressed numerically should be in commonly accepted, standardized units, and understood by the users of the device.	NA (The device does not have any diagnostic or measuring function.)		
2.5 Protection against radiation				
2.5.1	Devices should be designed and manufactured and packaged in such a way that exposure of patients, users and other persons to any emitted radiation should be reduced as far as practicable and appropriate, compatible with the intended purpose, whilst not restricting the application of appropriate specified levels for therapeutic and diagnostic purpose.	NA (The device does not have any radiation.)		
2.5.2	Intended radiation Where devices are designed to emit hazardous, or potentially hazardous, levels of visible and/or invisible radiation necessary for a specific medical	NA (The device does not have any radiation.)		

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	purpose the benefit of which is considered to outweigh the risks inherent in the emission, it should be possible for the user to control the emissions. Such devices should be designed and manufactured to ensure reproducibility of relevant variable parameters within an acceptable tolerance. Where devices are intended to emit potentially hazardous, visible and/or invisible radiation, they should be fitted, where practicable, with visual displays and/or audible warnings of such emissions.			
2.5.3	Unintended radiation Devices should be designed and manufactured in such a way that exposure of patients, users and other persons to the emission of unintended, stray or scattered radiation is reduced as far as practicable and appropriate.	NA (The device does not have any radiation.)		
2.5.4	Instructions for use The operating instructions for devices emitting radiation should give detailed information as to the nature of the emitted radiation, means of protecting	NA (The device does not have any radiation.)		

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	the patient and the user and on ways of avoiding misuse and of eliminating the risks inherent in installation.			
2.5.5	Ionizing radiation Devices intended to emit ionizing radiation should be designed and manufactured in such a way as to ensure that, where practicable, the quantity, geometry and energy distribution (or quality) of radiation emitted can be varied and controlled taking into account the intended use. Device emitting ionizing radiation intended for diagnostic radiology should be designed and manufactured in such a way as to achieve appropriate image and/or output quality for the intended medical purpose whilst minimizing radiation exposure of the patient and user. Devices emitting ionizing radiation, intended for therapeutic radiology should be designed and	NA (The device does not have any radiation.)		

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	manufactured in such a way as to enable reliable monitoring and control of the delivered dose, the beam type and energy and where appropriate the energy distribution of the radiation beam.			
2.6 Requirements for medical devices connected to or equipped with an energy source				
2.6.1	Devices incorporating electronic programmable systems, including software, should be designed to ensure the repeatability, reliability and performance of these systems according to the intended use. In the event of a single fault condition in the system, appropriate means should be adopted to eliminate or reduce as far as practicable and appropriate consequent risks.	A	ISO 13485:2016 ISO 14971:2019 IEC 60601-1:2005/A1:2012 /A2:2020 EN 60601-1-2:2015/A1:2021 IEC 60601-1-6:2010/A1:2013/A2:2020 IEC 80601-2-77:2019	Quality Manual (CX-001, Rev 22, 2025.03.01) Risk Management Report (310005, Rev 9, 2025.09.24) Compliance Test report - EMC Test Report (310403, Rev 3, 2025.08.07) - Safety Test Report (310404, Rev 3, 2024.07.22) Usability Evaluation Report

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			IEC TS 60601-4-2:2024 IEC 62366-1:2015/AMD1:2020 IEC 62304:2006/A1:2015	(310036, Rev 1, 2022.04.30) Software Validation Report (310027, Rev 12, 2025.09.30)
2.6.2	Devices where the safety of the patients depends on an internal power supply should be equipped with a means of determining the state of the power supply.	NA (The device is not intended for patient safety to depend on internal power supply)		
2.6.3	Devices where the safety of the patients depends on an external power supply should include an alarm system to signal any power failure.	NA (The device does not have the alarm system.)		

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2.6.4	Devices intended to monitor one or more clinical parameters of a patient should be equipped with appropriate alarm systems to alert the user of situations which could lead to death or severe deterioration of the patient's state of health.	NA (The device is not intended to monitor the clinical parameters of a patient)		
2.6.5	Devices should be designed and manufactured in such a way as to reduce as far as practicable and appropriate the risks of creating electromagnetic interference which could impair the operation of this or other devices or equipment in the usual environment.	A	IEC 60601-1:2005/A1:2012 /A2:2020 EN 60601-1-2:2015/A1:2021 IEC 60601-1-6:2010/A1:2013/A2:2020 IEC 80601-2-77:2019 IEC TS 60601-4-2:2024	Compliance Test report - EMC Test Report (310403, Rev 3, 2025.08.07) - Safety Test Report (310404, Rev 3, 2024.07.22) Risk Management Report (310005, Rev 9, 2025.09.24)

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			ISO 14971:2019	
2.6.6	Devices should be designed and manufactured in such a way as to provide an adequate level of intrinsic immunity to electromagnetic disturbance to enable them to operate as intended.	A	IEC 60601-1:2005/A1:2012 /A2:2020 EN 60601-1-2:2015/A1:2021 IEC 60601-1-6:2010/A1:2013/A2:2020 IEC 80601-2-77:2019 IEC TS 60601-4-2:2024 ISO 14971:2019	Compliance Test report - EMC Test Report (310403, Rev 3, 2025.08.07) - Safety Test Report (310404, Rev 3, 2024.07.22) Risk Management Report (310005, Rev 9, 2025.09.24)
2.6.7	Protection against electrical risks: Devices should be designed and manufactured in such a way as to avoid, as far as possible, the risk of accidental	A	IEC 60601-1:2005/A1:2012 /A2:2020	Compliance Test report - EMC Test Report (310403, Rev 3, 2025.08.07)

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	electric shocks during normal use and in single fault condition, provided the devices are installed and maintained as indicated by the manufacturer.		EN 60601-1-2:2015/A1:2021 IEC 60601-1-6:2010/A1:2013/A2:2020 IEC 80601-2-77:2019 IEC TS 60601-4-2:2024 ISO 14971:2019	- Safety Test Report (310404, Rev 3, 2024.07.22) Risk Management Report (310005, Rev 9, 2025.09.24)
2.7 Protection against mechanical risks				
2.7.1	Device should be designed and manufactured in such a way as to protect the patient and user against mechanical risks connected with, for example, resistance to movement, instability and moving parts.	A	IEC 60601-1:2005/A1:2012 /A2:2020 EN 60601-1-2:2015/A1:2021	Compliance Test report - EMC Test Report (310403, Rev 3, 2025.08.07) - Safety Test Report (310404, Rev 3, 2024.07.22)

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			IEC 60601-1-6:2010/A1:2013/A2:2020 IEC 80601-2-77:2019 IEC TS 60601-4-2:2024	
2.7.2	Devices should be designed and manufactured in such a way as to reduce to the lowest practicable level the risks arising from vibration generated by the devices, taking account of technical progress and of the means available for limiting vibrations, particularly at source, unless the vibrations are part of the specified performance.	A	IEC 60601-1:2005/A1:2012/A2:2020 EN 60601-1-2:2015/A1:2021 IEC 60601-1-6:2010/A1:2013/A2:2020 IEC 80601-2-77:2019 IEC TS 60601-4-2:2024	Compliance Test report - EMC Test Report (310403, Rev 3, 2025.08.07) - Safety Test Report (310404, Rev 3, 2024.07.22)

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2.7.3	Devices should be designed and manufactured in such a way as to reduce to the lowest practicable level the risks arising from the noise emitted, taking account of technical progress and of the means available to reduce noise, particularly at source, unless the noise emitted is part of the specified performance.	A	IEC 60601-1:2005/A1:2012 /A2:2020 EN 60601-1-2:2015/A1:2021 IEC 60601-1-6:2010/A1:2013/A2:2020 IEC 80601-2-77:2019 IEC TS 60601-4-2:2024	Compliance Test report - EMC Test Report (310403, Rev 3, 2025.08.07) - Safety Test Report (310404, Rev 3, 2024.07.22)
2.7.4	Terminals and connectors to the electricity, gas or hydraulic and pneumatic energy supplies which the user has to handle should be designed and constructed in such a way as to minimize all possible risks.	A	IEC 60601-1:2005/A1:2012 /A2:2020 EN 60601-1-2:2015/A1:2021	Compliance Test report - EMC Test Report (310403, Rev 3, 2025.08.07) - Safety Test Report (310404, Rev 3, 2024.07.22)

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			IEC 60601-1- 6:2010/A1:2013/A2:2020 IEC 80601-2-77:2019 IEC TS 60601-4-2:2024	
2.7.5	Accessible parts of the devices (excluding the parts or areas intended to supply heat or reach given temperatures) and their surroundings should not attain potentially dangerous temperatures under normal use.	A	IEC 60601- 1:2005/A1:2012 /A2:2020 EN 60601-1- 2:2015/A1:2021 IEC 60601-1- 6:2010/A1:2013/A2:2020 IEC 80601-2-77:2019 IEC TS 60601-4-2:2024	Compliance Test report - EMC Test Report (310403, Rev 3, 2025.08.07) - Safety Test Report (310404, Rev 3, 2024.07.22)
2.8 Protection against the risks posed to the patient by supplied energy or substances				

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2.8.1	Devices for supplying the patient with energy or substances should be designed and constructed in such a way that the delivered amount can be set and maintained accurately enough to guarantee the safety of the patient and of the user.	NA (The device is not intended to supply the patient with energy or substances)		
2.8.2	Devices should be fitted with the means of preventing and/or indicating any inadequacies in the delivered amount which could pose a danger. Devices should incorporate suitable means to prevent, as far as possible, the accidental release of dangerous levels of energy from an energy and/or substance source.	NA (The device is not intended to supply the patient with energy or substances)		
2.8.3	The function of the controls and indicators should be clearly specified on the devices. Where a device bears instruction required for its operation or indicates operating or adjustment parameters by means of a visual system, such information should	NA (The device is not intended to supply the patient with energy or		

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	be understandable to the user and, as appropriate, the patient.	substances)		
2.9 Protection against the risks posed to the patient for devices for self-testing or self-administration				
2.9.1	Such devices should be designed and manufactured in such a way that they perform appropriately for their intended purpose taking into account the skills and the means available to users and the influence resulting from variation that can reasonably be anticipated in user's technique and environment. The information and instructions provided by the manufacturer should be easy for the user to understand and apply.	A	IEC 60601-1:2005/A1:2012 /A2:2020 EN 60601-1-2:2015/A1:2021 IEC 60601-1-6:2010/A1:2013/A2:2020 IEC 80601-2-77:2019 IEC TS 60601-4-2:2024 EN 1041:2008 ISO 15223-1:2021	Compliance Test report - EMC Test Report (310403, Rev 3, 2025.08.07) - Safety Test Report (310404, Rev 3, 2024.07.22) User Manual (310023, Rev 15, 2025.10.15)

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2.9.2	Such devices should be designed and manufactured in such a way as to reduce as far as practicable the risk of use error in the handling of the device and, if applicable, the specimen, and also in the interpretation of results.	A	IEC 60601-1:2005/A1:2012 /A2:2020 EN 60601-1-2:2015/A1:2021 IEC 60601-1-6:2010/A1:2013/A2:2020 IEC 80601-2-77:2019 IEC TS 60601-4-2:2024 EN 1041:2008 ISO 15223-1:2021	Compliance Test report - EMC Test Report (310403, Rev 3, 2025.08.07) - Safety Test Report (310404, Rev 3, 2024.07.22) User Manual (310023, Rev 15, 2025.10.15)
2.9.3	Such devices should, where reasonably possible, include a procedure by which the user can verify	A	IEC 60601-1:2005/A1:2012 /A2:2020	Compliance Test report - EMC Test Report (310403, Rev 3, 2025.08.07)

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	that, at the time of use, which the product will perform as intended by the manufacturer.		EN 60601-1-2:2015/A1:2021 IEC 60601-1-6:2010/A1:2013/A2:2020 IEC 80601-2-77:2019 IEC TS 60601-4-2:2024 EN 1041:2008 ISO 15223-1:2021	- Safety Test Report (310404, Rev 3, 2024.07.22) User Manual (310023, Rev 15, 2025.10.15)
2.10 Information supplied by manufacturer				
2.10.1	Users should be provided with the information needed to identify the manufacturer, to use the device safely and to ensure the intended performance, taking account of their training and	A	EN 1041:2008 ISO 15223-1:2021	Labeling & Packaging Specification (310031, Rev 11, 2025.09.29)

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	knowledge. This information should be easily understood. NOTE 1: The requirements on this are addressed in the Sixth Schedule of the Medical Device Regulation.			User Manual (310023, Rev 15, 2025.10.15)
2.11 Performance evaluation including where appropriate, clinical evaluation				
2.11.1	All data generated in support of performance evaluation should be obtained in accordance with the relevant requirements applicable in each jurisdiction.	A	ASTM F2554-18 Internal Standard	Performance Test Report (Pose_Repeatability) (310053, Rev 0, 2021.11.19) Performance Test Report (Pose Accuracy) (310054, Rev 0, 2021.11.19) Performance Test Report (Workspace) (310055, Rev 0, 2021.11.19)
2.11.2	Clinical investigations on human subjects should be carried out in accordance with the spirit of the Helsinki Declaration. This includes every step in	NA (The device have not been		

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	the clinical investigation from first consideration of the need and justification of the study to publication of the results. In addition, some countries may have specific regulatory requirements for pre-study protocol review or informed consent. NOTE 2: Refer to the Third Schedule of Medical Device Regulation on Conformity Assessment Procedure and MDA/GD-xx: Clinical Evaluation on Medical Device for further information on the use of clinical evaluation to demonstrate compliance with these Essential Principle. This guidance document is in development process.	subjected to clinical investigation)		

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5. DESCRIPTION OF MEDICAL DEVICE

5.1. Detailed Description of the Medical Device Attribute

1) Device Description

The CUVIS-spine is a mobile system mainly comprising the robotic arm, the main console and the staff console as an option. The robotic arm is positioned on the floor near the side of the surgical table. The location of the main console or the staff console is appropriately determined considering the user preference and the environments.

2) Intended Use (Indications for use)

CS200 is intended for use as an aid for precisely locating anatomical structures and for the spatial positioning and orientation of guide bush to be used by surgeons for navigating and/or guiding compatible surgical instruments in open or percutaneous pedicle screw placement provided that the required markers and rigid patient anatomy can be identified on 3D or 2D image.

3) Product Specification

(1) System

- System: Robotic Arm, Main Console, Staff Console(Optional)
- Accessories (including options): Guide bush, Source Calibrator, Registration Tool, Registration Tool Adapter, Drape, Marker Ball, Patient Marker, Clamp, Adapter, Pin, Pin vise, Slide Hammer, Marker Driver, Detector calibrator, Dilator, Serration-tip dilator, Bur, Awl, Probe, Lenke probe, Tapper, Stylet tapper, Screwdriver, Instrument container, Handle.

(2) Electrical rating

- AC Power input: 100 – 240 VAC, 50/60Hz
- Power consumption: 1500VA (Robotic Arm), 280VA (Main Console), Staff 65VA (Staff Console)

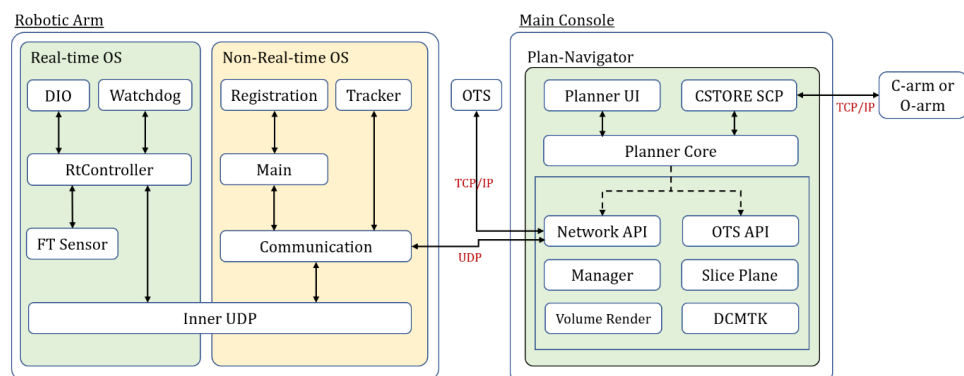
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(3) Software

① Software description

Category	Name	Function
Control Software	SRC	- User interface - Motion control
Planner Software	SPNe	- Image matching - Network with the robotic arm - Network with an imaging device - Network with OTS - GUI - DB management

② Software Component Architecture



- Robotic Arm

Component	Description
DIO	▪ To receive hardware button signal and transmit digital signals.
Watchdog	▪ To notify the abnormality of the robot controller by periodically exchanging signals with watchdog hardware module.

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Component	Description
RtController	To perform the real-time data processing and manipulator control with real-time OS module in real time cycle.
FT Sensor	To filter and convert the data acquired from the force torque sensor installed on the manipulator to the manipulator control signal.
Inner UDP	To communicate the data between real-time OS and non-real-time OS.
Registration	To perform the registration between the manipulator coordinate system and the image coordinate system, and the data transformation.
Tracker	To receive and filter the data
Main	To perform the calculation in non-real-time cycles, monitor the system status, and generate the control signals according to the system condition.
Communication	Network module of real-time OS area, Plan-Navigator, OTS.

- Main Console

Component	Description
Planner UI	Module that assist Graphical User Interface of Plan-Navigator.
Planner Core	Main operation module generally managing the two or three-dimensional data visualization function, motion management for each step, data processing of robotic arm and the other devices.
CSTORE SCP	Module that receives medical image data transmitted from medical imaging device by TCP / IP protocol with DICOM standard.
Network API	Ethernet network module. (supports TCP/IP, UDP)
OTS API	Optical Tracking System operation module.

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Component	Description
Manager	▪ Main operation module that manages the operation of Plan-Navigator for each step and the data processing with robotic arm and peripheral devices.
Volume Render	▪ Module for visualizing three-dimensional voxel data on two-dimensional screen.
DCMTK	▪ DICOM medical image processing module.
Imaging Device	▪ Medical imaging device that scans the target area and the registration tool, and transmits DICOM data to Plan-Navigator module.
OTS	▪ Optical Tracking system tracking three-dimensional position and orientation of the manipulator, the patient marker, the surgical instruments, etc.

(4) Operation, Transportation, and Storage condition

	Temperature	Humidity	Pressure
Operation	15~40℃	20~80%RH	80~106(kPa)
Transportation & Storage	0~40℃		

4) Configuration of the Device

No.	Part Description	Remark
1	Robotic Arm	System
2	Main Console	
3	Staff Console	
4	Guide bush	Accessory
5	Source Calibrator	
6	Registration Tool	
7	Registration Tool Adapter	
8	CUVIS-spine Drape	
9	Marker Ball	
10	Patient Marker	

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No.	Part Description	Remark
11	Clamp	
12	Marker Driver	
13	Detector Calibrator	
14	Dilator	
15	Serration-tip Dilator	
16	Bur	
17	Tapper	
18	Stylet Tapper	
19	Screwdriver	
20	Instrument Container	
21	Adapter	
22	Pin	
23	Pine Vise	
24	Slide Hammer	
25	Probe	
26	Awl	
27	Lenke Probe	
28	Handle	

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5.2. Complete Description of the Medical Device

1) System



[Robotic Arm]

[Main Console]

[Staff Console (option)]

No.	Name	Description
1	Robotic arm	- It consists of the manipulator, the end-effector and the base. - The manipulator controller is included in the base. - The manipulator moves to the goal pose according to the user's command.
2	Main console	- It consists of the OTS (optical tracking system) camera, OTS camera mounting arm, touch monitor, planner PC and mobile cart. - User can monitor the system status, and command the operation of the system through the main console.
3	Staff console (option)	- It consists of the touch monitor and the mobile cart. - It provides the same function as the main console while it is connected with the main console through the specified cables.

(1) Robotic arm

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No.	Name	Description
1	Manipulator	Component for moving the end-effector to goal position and orientation.
2	Manipulator brake release switch	Switch to release the motor brakes mounted on the 2nd joint and 3rd joint of the manipulator as required by the user. It is also possible to operate with the battery inside the base only when the battery residual level indicator on the port panel shows 24V or more. The battery inside the base is automatically charged when the external power source is connected to the robotic arm through the power cord supplied with the product.
3	LED (Orange)	A component to inform the user of the following conditions with visual information using light. States related to the movement of the manipulator such as movement coming soon, moving on, movement ending soon, etc. The operation of the lifting device.
4	End-effector	Component for guiding surgical instruments or implants.
5	End-effector marker	Marker needed for tracking the position and orientation of the end-effector by the optical tracking system.
6	Guide laser	Component for indicating target position by two line-lasers. Two line- lasers of Class 1 are built into the end-effector, and are turned ON or OFF by the [LASER/OTS] pedal.

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No.	Name	Description
7	Speaker	A component to inform the user of the following conditions with auditory information using sound. States related to the movement of the manipulator such as movement coming soon, moving on, movement ending soon, etc. The operation of the lifting device.
8	Handle	Area that the user holds by hand when moving the robotic arm.
9	Base	Component that supports the manipulator, including casters for the movement of robotic arm, lifting device for fixing to the floor, manipulator controller, and power supplies.
10	Caster	Device for transporting the robotic arm. Totally locked and unlocked by using the brake pedal and the brake release pedal, respectively.
11	Lifting device	Device for fixing the base to the floor for the stability of the robotic arm during the operation of the manipulator. There are 4 lift actuators on the bottom of the base. They are operated by the lift switch on the control panel, and driven by the lift control box inside the base. It is also possible to operate with the battery inside the base only when the battery residual level indicator on the port panel shows 24V or more.
12	Intake hole	Holes for intake of outside air for cooling inside the base.
13	Robot marker	Marker used to estimate the reference coordinate system of the manipulator by the optical tracking system. The relative position and orientation between the robot marker and the reference coordinate system of the manipulator can be obtained through the process of the manipulator calibration.
14	Control panel	Panel with emergency stop switch, power switch, initialization switches and lift switch
15	Port panel	Panel with HDMI port for connecting external monitor, USB ports, foot switch connection port, battery residual level indicator, Ethernet port and ELCB.
16	Ventilating hole	Holes for ventilating heat generated inside the base to the outside.
17	Power inlet	AC power inlet
18	Brake pedal	Pedal for locking the front and rear casters.
19	Brake release pedal	Pedal for unlocking the front and rear casters.

① Port panel

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No.	Name	Description
1	HDMI port	Port for connecting to an external monitor.
2	USB port	Ports for USB memory, keyboard, or mouse. (not included)
3	Foot switch connection port	Port for connecting to the foot switch.
4	Battery residual level indicator	Device for indicating the residual level of the battery as percentage or voltage. Press the power button on the battery residual level indicator to have it display the residual level as percentage. When pressed again, the residual voltage is displayed.
5	Ethernet port	Port for the network cable to communicate with the main console.
6	ELCB (Earth leakage circuit breaker)	Device for cutting off power when detecting abnormal conditions such as a short circuit, overload or current leakage.

② Control panel

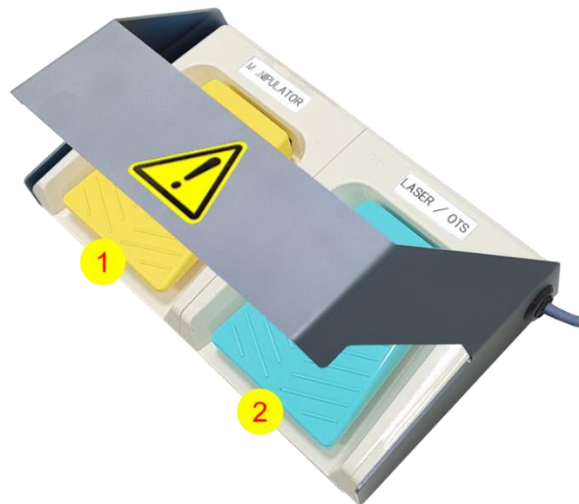
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No.	Name	Description
1	Emergency stop switch	Switch for emergency stop of manipulator movement. When pushing emergency stop switch, the movement of manipulator stops and motors are powered off immediately. At that time, red LED is on. To release emergency state, turn the emergency switch clockwise and initialization of robotic arm perform again.
2	Power switch	Power on / off switch of the robotic arm.
3	Initialization switch (left)	Manipulator initialization button when the robotic arm is placed on the left side of the operating table.
4	Initialization switch (right)	Manipulator initialization button when the robotic arm is placed on the right side of the operating table.
5	Lift switch	Switch for operating the lifting device of the base. [DOWN] to raise the base for fixing the robotic arm to the floor. [UP] to lower the base for releasing the robotic arm. Middle position to stop operating the lifting device.

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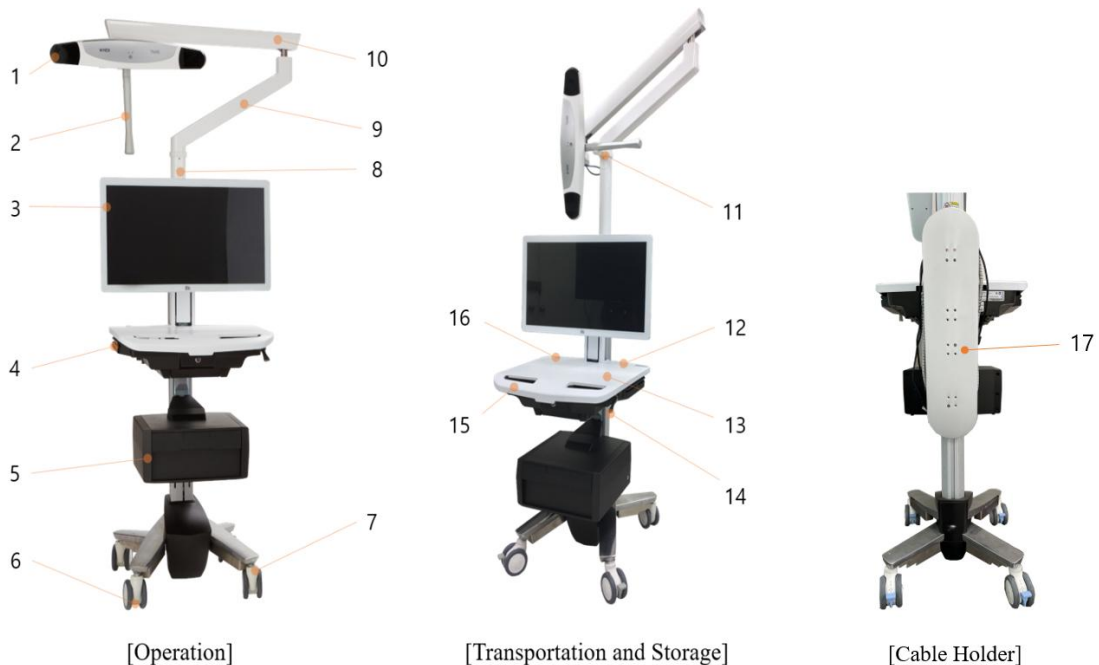
③ Foot switch



No.	Name	Description
1	Manipulator	Switch to start, pause or resume the movement of the manipulator. The manipulator starts or continues moving only while the user presses the [MANIPULATOR] pedal. When the user releases the pedal, the manipulator stops moving, and when the user presses it again, it resumes the movement.
2	Laser / OTS	Switch to turn on or off the guide laser of the end-effector, on and off state is changed after the pedal is pressed and released. Switch to acquire tracking data from the optical tracking system in the step of SCAN.

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(2) Main console



No.	Name	Description
1	OTS camera	Camera of the optical tracking system. If initialization fails, touch OTS icon in the system status area at the bottom right of the touch screen monitor to retry initialization.
2	OTS camera handle	Handle that the user holds to change the position and orientation of the OTS camera.
3	Touch monitor	Touch screen monitor for the surgical user interface.
4	Main power and Port panel	Main power of main console. Composed of VGA port, Ethernet ports and USB port.
5	Drawer	Space for storing accessory parts including cables. (Maximum work load: 1kg)
6	Caster lock pedal	Pedals for locking the casters to stop moving the main console.
7	Caster	Castors supporting the main console and providing mobility.
8	Bottom body	Supports OTS camera and mounting arm, and includes parts for connecting the mounting arm and the mobile cart.

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No.	Name	Description
9	Lower arm	Supports the upper Arm of OTS mounting arm.
10	Upper arm	Equipped with joints for adjusting the height and direction of the OTS camera.
11	Parking clamp	Clamp that secures the OTS camera handle to prevent the upper arm from moving during transportation or storage.
12	LED (Blue)	Indicates On / off status of the planner PC inside the storage. When the planner PC is on, the LED(Blue) is on. When the planner PC is off, the LED(Blue) is off.
13	Storage	Space that stores planner PC, network hub, terminal block, power adapters, etc. Opening is prohibited except for the manufacturer or the person authorized by the manufacturer.
14	Height adjustment brake handle	Handle for adjusting or locking the height of the storage and the touch monitor. The relative height of the touch monitor with respect to the storage cannot be changed.
15	Handle	Area that the user holds for catching or moving the main console.
16	Power switch	Switch for only turning on the planner PC by inputting AC power to the planner PC in the storage. (not to be used to turn off planner PC) Switch for powering on / off all equipment in the storage. Indicates On / off status of the main power. When the main power is on, the LED(Blue) is on. When the main power is off, the LED(Blue) is off.
17	Cable holder	Holder for storing HDMI/USB cables that connect to the staff console

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① Port panel



No.	Name	Description
1	Power Inlet	AC power inlet
2	VGA port	Display port for the connection with the staff console or other display device.
3	Ethernet port	Port for the network cables connecting with the robotic arm and the intraoperative imaging device.
4	USB port	Port for the touch signal cable of the staff console, USB memory, keyboard, or mouse. (not included)

② Control panel



No.	Name	Description
1	Power switch	Switch for only turning on the planner PC by inputting AC power to the planner PC in the storage. (not to be used to turn off planner PC) Switch for powering on / off all equipment in the storage. Indicates on / off status of the main power. • When the main power is on, the LED(Blue) is on. • When the main power is off, the LED(Blue) is off.
2	Planner PC	Indicates on / off status of the planner PC inside the storage.

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No.	Name	Description
	Power Status LED (Blue)	- When the planner PC is on, the LED(Blue) is on - When the planner PC is off, the LED(Blue) is off.

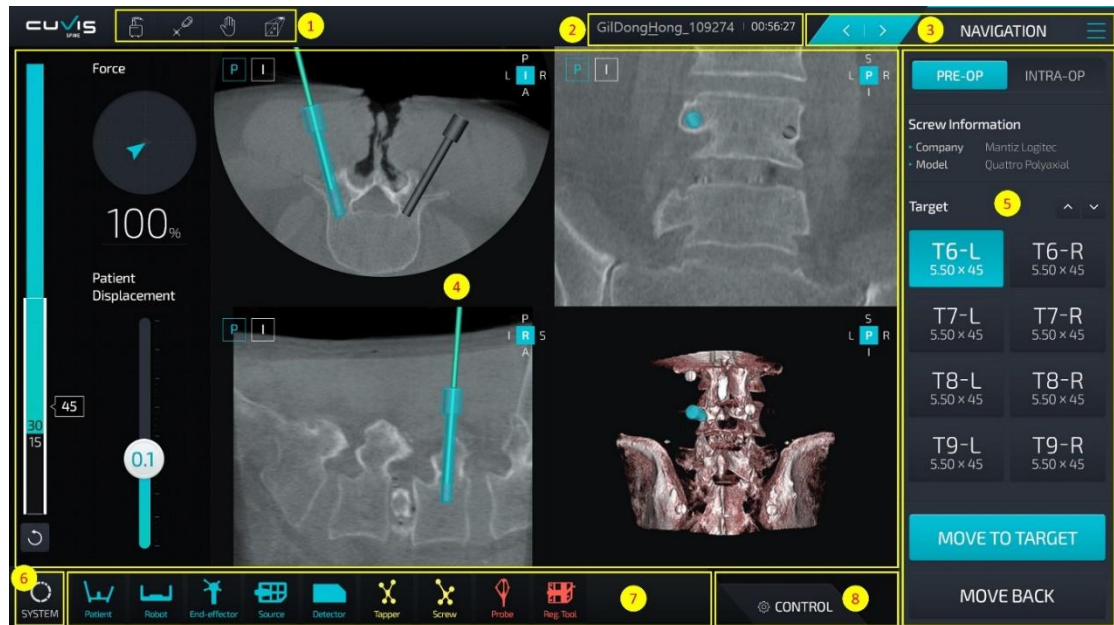
(3) Staff console (option)



No.	Name	Description
1	Touch monitor	Touch screen monitor for the surgical user interface.
2	Storage	Space for storing one USB cable and one VGA cable to connect with the main console. Space for storing power cord.
3	Handle	Area that the user holds for moving or catching the staff console.
4	Caster lock pedal	Pedals for locking the casters to stop moving the staff console.
5	Caster	Casters supporting the staff console and providing mobility.
6	Height adjustment brake handle	Handle for adjusting or locking the height of the storage and the touch monitor. The relative height of the touch monitor with respect to the storage cannot be changed.
7	Monitor power switch	Switch to manually turn on or off the touch screen monitor When the external power is supplied, it is in standby mode unless it is turned off by the power switch on purpose. When a video signal is received from the main console, the touch screen monitor is turned on.
8	Power Inlet	AC power inlet

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(4) Software user interface



① Hotkey

These hotkeys are accessible and active at all steps, and the function of each button is shown below. They are not displayed when pre-op. 3D is selected as imaging modality.

Key Status		Name	Description
Off	On		
		Move to ready	Move the manipulator to the ready position.
		Guide laser	Turn on or off the guide laser.
		Hand guide	Activate the hand guide function, and display a pop-up window.
		Marker ball view	The top view and side view of the OTS tracking volume are displayed, and the positions of the tracked markers are displayed in each view.

② Patient info and Time

The patient information and elapsed time are displayed.

③ Step

Allows the user to access the top level of each step.
Proceeds as the surgery progresses.

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Each tab is accessible at any time.

④ Main working area

Provides information, displays images or allows user input.

⑤ Function menu

Selects information or give a command to the device.

⑥ System status icon

It displays the connection status between the main console and system components and whether they are operating normally.

If the system components are normally connected to the main console and are operating normally, the text and picture of the system status icon are displayed in white.

If the system components are normally connected to the main console, but any one of them is not operating normally, the text and picture of the system status icon are displayed in yellow. In this case, check the status of each component by touching the system status icon, and take actions according to the pop-up instructions by touching the component icon displayed in yellow as shown below.

If the system components are not properly connected to the main console or an error has occurred, the text and picture of the system status icon are displayed in red. In this case, touch the system status icon to check the status of each component, check the cable connection and take actions according to the pop-up instructions by touching the component icon displayed in red as shown in below.

Component status icons			Name	Description
Normal	Warning	Error		
			OTS	Optical tracking system (Polaris Vega; NDI co.)
			Robot	Robotic arm
			2D imaging device	Intra-op. 2D imaging device (e.g. C-arm)

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			3D imaging device	Intra-op. 3D imaging device (e.g. O-arm, 3D C-arm)
			Registration Module	Indicates the connection status of the 2D-3D image registration module

The component status icons are not displayed when pre-op. 3D is selected as imaging modality.

⑦ Marker status area

Indicates whether the markers are being tracked.

Markers are displayed in green or yellow when tracking is possible using the optical tracking system. On the other hand, if tracking is impossible, it is displayed in red.

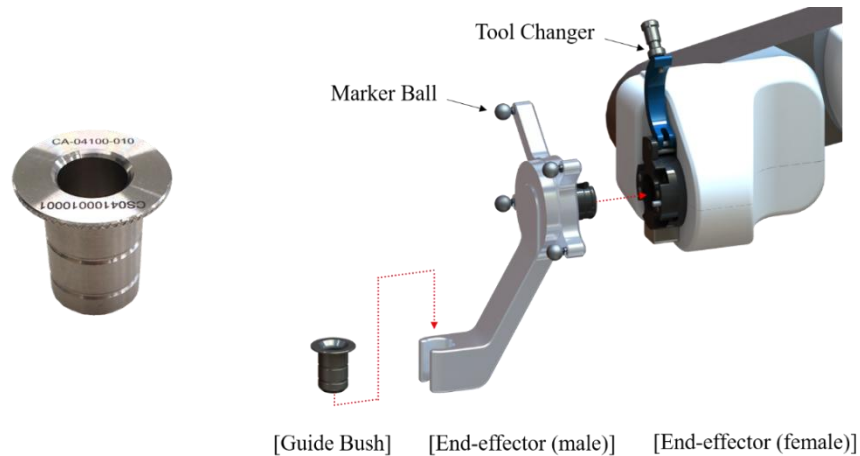
⑧ Control menu

Controls objects displayed in the main working area, those are selectively activated in Scan, Planning or Navigation tabs.

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2) Accessory

① Guide Bush



No.	Model Number	Description	Sterilization/ Disposable
1	GB-0950	It is temporarily mounted on the end-effector when the surgical instrument or screw is guided and supported by the manipulator	User Steam Sterilization
2	GB-1250		
3	GB-1300		
4	GB-1350		
5	GB-1400		
6	GB-1450		
7	GB-1500		
8	GB-1550		
9	GB-1600		
10	GB-0953		
11	GB-1199		
12	GB-1303		
13	GB-1356		
14	GB-1373		
15	GB-1410		
16	GB-1388		
17	GB-1398		
18	GB-1510		
19	GB-1613		

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② Source Calibrator



No.	Model Number	Description	Sterilization/ Disposable
1	SC-001	It is temporarily attached to the registration tool adapter mounted on the end-effector during 2D scan. It must be covered with sterile tool drape before attaching to the registration tool adapter.	Non-sterile part

③ Registration Tool



RT-0100 (left) and RT-0200 (right)

No.	Model Number	Description	Sterilization/ Disposable
1	RT-001	It is temporarily attached to the registration tool adapter mounted on the end-effector during 3D scan. It	Non-sterile part

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No.	Model Number	Description	Sterilization/ Disposable
2	RT-002	must be covered with sterile tool drape before attaching to the registration tool adapter.	

④ Registration Tool Adaptor



No.	Model Number	Description	Sterilization/ Disposable
1	RA-001	It is temporarily mounted on the end-effector when the registration tool (for intra-op 3D scan) or the source calibrator (for intra-op 2D scan) has to be connected with the end-effector.	User Steam Sterilization

⑤ CUVIS-spine Drape



Robotic Arm Drape



Tool Drape



Detector Drape

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No.	Model Number	Description	Sterilization/ Disposable
1	Robotic Arm Drape	It is used for covering the robotic arm with sterilized vinyl drape in sterile zone.	Disposable, EO gas sterilization
2	Tool Drape	It is used for covering the source calibrator or the registration tool with the sterilized vinyl drape in sterile zone.	
3	Detector Drape	It is used for covering the detector calibrator with the sterilized vinyl drape in sterile zone.	

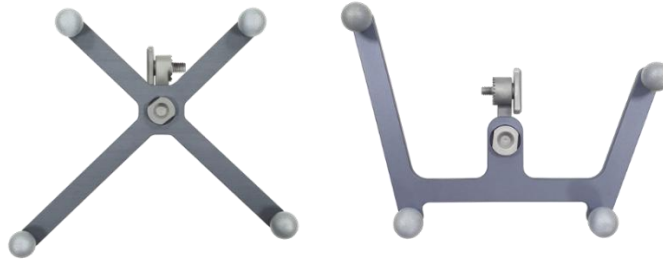
⑥ Marker ball



No.	Model Number	Description	Sterilization/ Disposable
1	MB-001	A disposable infrared reflective marker sphere used to obtain position data using OTS.	Disposable, EO gas sterilization
2	MB-002		

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⑦ Patient marker



PM-F001 (left) and PM-F002 (right)

No.	Model Number	Description	Sterilization/ Disposable
1	PM-F001	It is used to track the movement of the vertebra body in real time.	User Steam Sterilization
2	PM-F002		

⑧ Clamp



Straight Type and Curved Type

No.	Model Number	Type	Description	Sterilization/ Disposable
1	PM-C001	Straight, Short	It is used to fix the patient marker to the spinous process. It is used in combination with the patient marker.	User Steam Sterilization
2	PM-C002	Straight, Long		
3	PM-C011	Curved, Short		
4	PM-C012	Curved, Medium		
5	PM-C013	Curved, Long		

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⑨ Adapter



No.	Model Number	Description	Sterilization/ Disposable
1	PM-A001	It is used in combination with the patient marker and the pin.	User Steam Sterilization

⑩ Pin



No.	Model Number	Description	Sterilization/ Disposable
1	PM-P100	It is a part inserted into the iliac crest to fix the patient marker. It is used in combination with the patient marker (PM-F002) and the adapter (PM-A001)	User Steam Sterilization
2	PM-P150		
3	PM-P200		

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⑪ Pin vise



No.	Model Number	Description	Sterilization/ Disposable
1	PV-0001	It is used to insert the pin into the bone with a mallet.	User Steam Sterilization

⑫ Slide hammer



No.	Model Number	Description	Sterilization/ Disposable
41	SH-0001	It is used to remove the pin that is fixed to the bone.	User Steam Sterilization

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⑬ Marker driver



No.	Model Number	Description	Sterilization/ Disposable
1	MD-0000	It is used in the following cases. - To fix or release the clamp to or from the bone. - To connect or separate the patient marker and the clamp or the adapter. - To connect or separate the adapter and the pin.	User Steam Sterilization

⑭ Detector calibrator



DC-GM001 (upper left), DC-GE001 (upper right),
DC-SM001 (lower left) and DC-DK001 (lower middle), DLDC-U001(lower right)

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No.	Model Number	Detector type	Description	Sterilization/ Disposable
1	DC-GE001	Flat panel	A part that is mounted on detector of 2D imaging device.	Non-sterile part
2	DC-GM001	Image intensifier		
3	DC-SM001			
4	DC-DK001			
5	DLDC-U001			

⑮ Dilator



No.	Model Number	Description	Sterilization/ Disposable
1	DL-1300	It is used to secure the insertion path of other surgical instruments, and composed of the shaft and the cannula that can be combined or separated	User Steam Sterilization
2	DL-1775		
3	DL-1300-MSZ		
4	DL-1775-MSZ		
5	DL-1300-MSI		
6	DL-1300-LKP		
7	DL-1775-LKP		

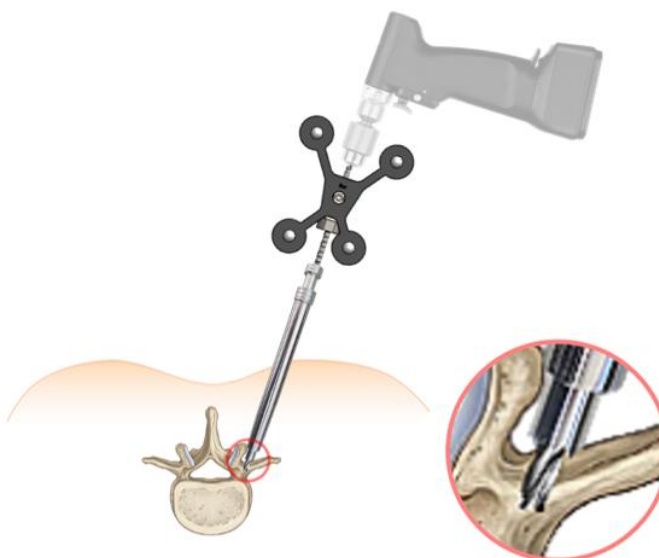
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16 Serration-tip dilator



No.	Model Number	Description	Sterilization/ Disposable
1	SD-0500	It is used to keep the position of the bur stable	User Steam Sterilization
2	SD-2030		
3	SD-1630BA		
4	SD-2030BA		

17 Bur



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a. Bur Marker



BM-0800(left), CM1325(middle) and BAM(right)

No.	Model Number	Description	Sterilization/ Disposable
1	BM-0800	It is assembled with a compatible model of bur.	User Steam Sterilization
2	CM-1325		
3	BAM		

b. Bur



No.	Model name	Head diameter (mm)	Length (mm)	Shaft diameter (mm)	Shank type	Description	Sterilization/ Disposable
1	MR223-30BE	3.0	223	2.35	Midas Rex Legend	It is used in conjunction with electrically-powered drill handpiece to make a pilot hole in the bone.	Disposable, User Steam Sterilization
2	AN223-30BE			2.35	Anspach		
3	ST223-30BE			1.56	Stryker		
4	LI223-30BE			3.16	Linvatec		
5	AO223-30BE			4.48	AO Synthes		

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No.	Model name	Head diameter (mm)	Length (mm)	Shaft diameter (mm)	Shank type	Description	Sterilization/ Disposable
6	MR227-30BE		277	2.35	Midas Rex Legend		
7	AOL300-30BE*		300	5.00	AO Synthes		
8	NS255-30BE		255	2.35	NSK		
9	M8238-30BE	3.0	238	2.35	Midas Rex M8		
10	M8277-30BE		277	2.35	Midas Rex M8		
11	AO277-30BE		277	2.35	AO Synthes		
12	AC232-30BE		232	2.35	AESCU LAP		
13	BA232-30BE		232	2.35	Bienair		
14	BA274-30BE		274	2.35	Bienair		
15	MR223-40BE	4.0	223	2.35	Midas Rex Legend		
16	AN223-40BE			2.35	Anspach		
17	ST223-40BE			1.56	Stryker		
18	LI223-40BE			3.16	Linvatec		
19	AO223-40BE			4.48	AO Synthes		
20	MR277-40BE		277	2.35	Midas Rex Legend		
21	AOL300-40BE*		300	5.00	AO Synthes		
22	NS255-30BE		255	2.35	NSK		
23	M8238-30BE		238	2.35	Midas Rex M8		
24	M8277-30BE		277	2.35	Midas Rex M8		
25	AO277-30BE		277	2.35	AO Synthes		
26	AC232-30BE		232	2.35	AESCU LAP		

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No.	Model name	Head diameter (mm)	Length (mm)	Shaft diameter (mm)	Shank type	Description	Sterilization/ Disposable
27	BA232-30BE		232	2.35	Bienair		
28	BA274-40BE		274	2.35	Bienair		

18 Awl



a. Sharp Awl Marker



No.	Model Number	Description	Sterilization/ Disposable
1	SAM-0800	It is assembled with a Sharp Awl Body.	User Steam Sterilization

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b. Sharp Awl Body



No.	Model Number	Description	Sterilization/ Disposable
1	AW-0000	It is used to pierce cortical bone.	User Steam Sterilization

19 Probe



No.	Model Number	Description	Sterilization/ Disposable
1	PB-0000	It is used to check the point where you want to check the location.	User Steam Sterilization
2	PB-0003		

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20 Lenke probe



a. Lenke Straight Probe Marker



No.	Model Number	Description	Sterilization/ Disposable
1	LSPM-0800	It is assembled with a Lenke Straight Probe Body	User Steam Sterilization

b. Lenke Straight Probe Body



No.	Model Number	Description	Sterilization/ Disposable
1	PB-0001	If calibration is completed, the real-time location of the lenke probe tip is displayed on the navigation screen	User Steam Sterilization

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21 Tapper



a. Tapper Marker



No.	Model Number	Description	Sterilization/ Disposable
1	TM-0950	It is assembled with a compatible model of Tapper Body or Stylet	User Steam Sterilization
2	TM-1350		
3	TM-0950S	Tapper Body	

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b. Tapper Body



No.	Model Number	Head diameter (mm; nominal value)	Description	Sterilization/ Disposable
1	NT-0300	Ø3.0	It is used to make a hole and Threads.	User Steam Sterilization
2	NT-0350	Ø3.5		
3	NT-0400	Ø4.0		
4	NT-0450	Ø4.5		
5	NT-0500	Ø5.0		
6	NT-0550	Ø5.5		
7	NT-0600	Ø6.0		
8	NT-0650	Ø6.5		
9	NT-MS45	Ø4.5		
10	NT-MS55	Ø5.5		
11	NT-MS65	Ø6.5		
12	NT-MS75	Ø7.5		
13	CT-MS45	Ø4.5		
14	CT-MS55	Ø5.5		
15	CT-MS65	Ø6.5		
16	CT-MS75	Ø7.5		
17	NT-LK45	Ø4.5		
18	NT-LK55	Ø5.5		
19	NT-LK65	Ø6.5		
20	CT-LK45	Ø4.5		
21	CT-LK55	Ø5.5		
22	CT-LK65	Ø6.5		
23	NT-LK45P*	Ø4.5		
24	NT-LK55P*	Ø5.5		
25	NT-LK65P*	Ø6.5		

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22 Stylet Tapper

*Assemble with Tapper Marker



a. Stylet Tapper Body



No.	Model Number	Head Diameter (mm; nominal value)	Description	Sterilization/ Disposable
1	ST-0550	Φ5.5	It is used to make a hole and Threads. Include a stylet for anchoring on the bone surface and the stylet lever for its operation.	User Steam Sterilization
2	ST-0600	Φ6.0		
3	ST-0650	Φ6.5		

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23 Screwdriver



a. Screwdriver Marker



No.	Model Number	Description	Sterilization/ Disposable
1	SM-0700	It is assembled with a compatible model of Screwdriver Body	User Steam Sterilization
2	SM-0800		
3	SM-0850		
4	SM-1000		
5	SM-1200		
6	SM-U001		
7	SM-0800S		
8	SM-1200S		

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b. Screwdriver Body



MQ-0000(left), UI-0000(middle) and LK-XXXX(right)

No.	Model Number	Description	Sterilization/ Disposable
1	LK-0001	It is used to insert a screw into a threaded hole	User Steam Sterilization
2	LK-0002		
3	LK-0003		

24 Instrument container



No.	Model Number	Height (mm; nominal value)	Description	Sterilization/ Disposable
1	CC-0000	204	It is used for storing the cleaning and sterilization surgical instruments	User Steam Sterilization
2	CM-0000	154		

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25 Handle



A1A-1047 (left), LS03-0304 (right)

No.	Model Number	Description	Sterilization/ Disposable
1	A1A-1047	It is assembled with compatible model of handle use accessories	User Steam Sterilization
2	LS03-0304		

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5.3. Principles of Operation

CUVIS-spine is a pedicle screw guide system used as a navigation to guide surgical tools to the user's planned target position and posture in incision or percutaneous spinal surgery, and consists of the robotic arm, main console, and staff console (option).

The coordinate system matching between the image and the device is achieved by fixing the patient marker to the bone adjacent to the surgical site, and photographing the surgical site of the patient with an imaging device together with the matching tool mounted on the end effector. The user makes a surgical plan while watching the video, and when the device is ordered to move, the end effector moves to guide the surgical tool. Movement of the vertebrae that may occur during surgery can be corrected the guide position by tracking the patient marker.

5.4. Risk Class and Applicable Classification Rule

CUVIS-spine is classified as a Class B medical device in accordance with the medical device classification system stipulated under Decree No. 98/2021/ND-CP of Vietnam, which is based on the ASEAN Medical Device Directive (AMDD).

5.5. Description of the accessories, other medical devices and other products that are not medical devices, which are intended to be used in combination with the medical device.

- 1) Compatible components with the system (Robotic Arm, Main Console, Staff Console (Option))

Part Name	Model No.	Type	Length	Manufacturer	Qty.
Power Cable	6051.2123	CEE7/VII	6m	Schurter	3
	3-102-524	NEMA 5-15P	6m	Schurter	3
	IL19-EU1-H05-3150-200	CEE7/VII	2m	Schaffner	3
	IEC-G16B-1.5	CEE7/VII	2m	DONGYANG	3
Ethernet Cable	NM-U710Z	-	10m	Gangwon Electronic	2
USB Cable	U-10374	-	10m	UGREEN	1
HDMI Cable	BWHFOC	-	10m	ARTMU	1
DP to HDMI	DHHC100	-	1m	ARTMU	1

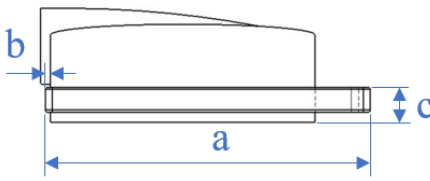
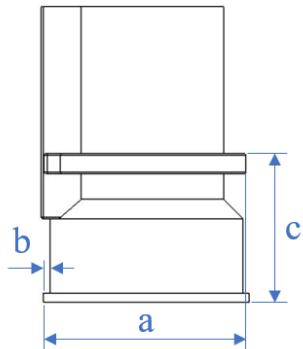
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Part Name	Model No.	Type	Length	Manufacturer	Qty.
Cable					
Ferrite Core	HT3329T	-	-	TODAISU	5
Auxiliary parts	Mouse	USB(Wireless)	-	-	1
	Keyboard	USB(Wireless)	-	-	1
	CD-ROM	USB	-	-	1

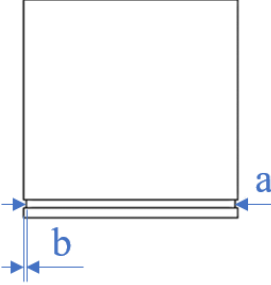
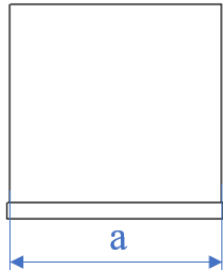
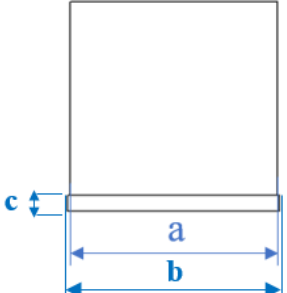
2) Compatible imaging device specification with Detector calibrator

No.	Detector calibrator	Mobile fluoroscopic system	
	Model name	Manufacturer	Model name
1	DC-GM001	GEMSS Medical	SPINEL 12HD
2	DC-GE001	GE Healthcare	OEC 9900 Elite
3			OEC 8800 FlexView
4	DC-SM001	Siemens Healthcare	Arcadis Orbic Eco
5	DC-DK001	DK Medical Systems	PROSTAR
6	DLDC-U001	GE Healthcare	OEC 8800 FlexView
		DK Medical Systems	PROSTAR

[Shape and size of compatible detector by model]

Model name	Shape	a	b	c
DC-GM001		470 or less	6.5~11	48~51
DC-GE001		426 or less	7 or more	168~264

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Model name	Shape	a	b	c
DC-SM001		274~312	3~8	-
DC-DK001		284~320	-	-
DLDC-U001		285~350	280~325	5~45

3) Compatible Bienair handpiece specification with Bur Marker

No.	Bur Marker	handpiece	
	Model Name	Company	Model
1	BAM	Bienair	PM2-S150

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5.6. Description or complete list of the various configurations of the medical device to be registered

1) System

No.	Name	Description
1	Robotic Arm (Option)	- It consists of the manipulator, the end-effector and the base. - The manipulator controller is included in the base. - The manipulator moves to the goal pose according to the user's command.
2	Main Console	- It consists of the OTS (optical tracking system) camera, OTS camera mounting arm, touch monitor, planner PC and mobile cart. - The software that supports surgical navigation runs by default. - When connected to the robotic arm, the user can monitor the system status and issue operational commands through the main console.
3	Staff Console (Option)	- It consists of the touch monitor and the mobile cart. - It provides the same function as the main console while it is connected with the main console through the specified cables.

2) Accessory

No.	Name	Model Number	Description	Note (Sterilization/Disposable)
1	Guide Bush	GB-0950	It is temporarily mounted on the end-effector when the surgical instrument or screw is guided and supported by the manipulator	User Steam Sterilization
2		GB-1250		
3		GB-1300		
4		GB-1350		
5		GB-1400		
6		GB-1450		

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No.	Name	Model Number	Description	Note (Sterilization/Disposable)
7		GB-1500		
8		GB-1550		
9		GB-1600		
10		GB-0953		
11		GB-1199		
12		GB-1303		
13		GB-1356		
14		GB-1373		
15		GB-1410		
16		GB-1388		
17		GB-1398		
18		GB-1510		
19		GB-1613		
20	Scalpel Gude	SG-1350	It is temporarily mounted on the end-effector when the Scalpel Handle is guided and supported by the manipulator (User Steam Sterilization)	User Steam Sterilization
21	Scalpel Handle	SH-1350	It is used in combination with a scalpel when incising points guided by a manipulator. (User Steam Sterilization)	User Steam Sterilization
22	Source Calibrator	SC-001	It is temporarily attached to the registration tool adapter mounted on the end-effector during 2D scan. It must be covered with sterile tool drape before attaching to the registration tool adapter.	Non-sterile part
23	Registration Tool	RT-001	It is temporarily attached to the registration tool adapter mounted on the end-effector during 3D scan. It must be covered with sterile tool drape before attaching to the registration tool adapter.	Non-sterile part
24		RT-002		

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No.	Name	Model Number	Description	Note (Sterilization/Disposable)
25	Registration Tool Adaptor	RA-001	It is temporarily mounted on the end-effector when the registration tool (for intra-op 3D scan) or the source calibrator (for intra-op 2D scan) has to be connected with the end-effector.	User Steam Sterilization
26	Robotic Arm Drape	RBC001	It is used for covering the robotic arm with sterilized vinyl drape in sterile zone.	Disposable, EO gas sterilization
27	Tool Drape	RBC004C	It is used for covering the source calibrator or the registration tool with the sterilized vinyl drape in sterile zone.	
28	Detector Drape	RBC005C	It is used for covering the detector calibrator with the sterilized vinyl drape in sterile zone.	
29	Marker Ball (Alternative)	MB-001	A disposable infrared reflective marker sphere used to obtain position data using OTS.	Disposable, EO gas sterilization
30		MB-002		
31	Patient Marker	PM-F001	It is used to track the movement of the vertebra body in real time.	User Steam Sterilization
32		PM-F002		
33	Clamp	PM-C001	It is used to fix the patient marker to the spinous process.§ It is used in combination with the patient marker.	User Steam Sterilization
34		PM-C002		
35		PM-C011		
36		PM-C012		
37		PM-C013		
38	Adaptor	PM-A001	It is used in combination with the patient marker and the pin.	User Steam Sterilization
39	Pin	PM-P100	It is a part inserted into the iliac crest to fix the patient marker. It is used in combination with the patient marker (PM-F002) and the adapter (PM-A001)	User Steam Sterilization
40		PM-P150		
41		PM-P200		
42	Pin Vice	PV-0001	It is used to insert the pin into the bone with a mallet.	User Steam Sterilization

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No.	Name	Model Number	Description	Note (Sterilization/Disposable)
43	Slide Hammer	SH-0001	It is used to remove the pin that is fixed to the bone.	User Steam Sterilization
44	Marker Driver	MD-0000	It is used in the following cases. - To fix or release the clamp to or from the bone. - To connect or separate the patient marker and the clamp or the adapter. - To connect or separate the adapter and the pin.	User Steam Sterilization
45	Detector Calibrator	DC-GE001	A part that is mounted on detector of 2D imaging device.	Non-sterile part
46		DC-GM001		
47		DC-SM001		
48		DC-DK001		
49		DLDC-U001		
50	Dilator	DL-1300	It is used to secure the insertion path of other surgical instruments, and composed of the shaft and the cannula that can be combined or separated	User Steam Sterilization
51		DL-1775		
52		DL-1300-MSZ		
53		DL-1775-MSZ		
54		DL-1300-MSI		
55		DL-1300-LKP		
56		DL-1775-LKP		
57	Serration-tip Dilator	SD-0500	It is used to keep the position of the bur stable	User Steam Sterilization
58		SD-2030		
59		SD-1630BA		
60		SD-2030BA		

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No.	Name	Model Number	Description	Note (Sterilization/Disposable)
61	Bur Marker	BM-0800	It is assembled with a compatible model of bur.	User Steam Sterilization
62		CM-1325		
63		BAM		
64	Bur	MR223-30BE	It is used in conjunction with electrically-powered drill handpiece to make a pilot hole in the bone.	Disposable, User Steam Sterilization
65		AN223-30BE		
66		ST223-30BE		
67		LI223-30BE		
68		AO223-30BE		
69		MR223-40BE		
70		AN223-40BE		
71		ST223-40BE		
72		LI223-40BE		
73		AO223-40BE		
74		MR277-30BE		
75		MR277-40BE		
76		AOL300-30BE		
77		AOL300-40BE		
78		M8238-30BE		
79		M8238-40BE		
80		M8277-30BE		
81		M8277-40BE		

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No.	Name	Model Number	Description	Note (Sterilization/Disposable)
82		BA232-30BE		
83		BA232-40BE		
84		AO277-30BE		
85		AO277-40BE		
86		NS255-30BE		
87		NS255-40BE		
88		AC232-30BE		
89		AC232-40BE		
90		BA274-30BE		
91		BA274-40BE		
92	Sharp Awl Marker	SAM-0800	It is assembled with a Sharp Awl Body.	User Steam Sterilization
93	Sharp Awl Body	AW-0000	It is used to pierce cortical bone.	User Steam Sterilization
94	Probe	PB-0000	It is used to check the point where you want to check the location.	User Steam Sterilization
95		PB-0003		
96	Lenke Straight Probe Marker	LSPM-0800	It is assembled with a Lenke Straight Probe Body	User Steam Sterilization
97	Lenke Straight Probe Body	PB-0001	If calibration is completed, the real-time location of the lenke probe tip is displayed on the navigation screen	User Steam Sterilization
98	Tapper Marker	TM-0950	It is assembled with a compatible model of Tapper Body or Stylet Tapper Body	User Steam Sterilization
99		TM-1350		
100		TM-0950S		
101	Stylet Tapper Body	ST-0550	It is used to make a hole and Threads. Include a stylet for anchoring on the	User Steam Sterilization

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No.	Name	Model Number	Description	Note (Sterilization/Disposable)
102		ST-0600	bone surface and the stylet lever for its operation.	
103		ST-0650		
104	Tapper Body	NT-0300	It is used to make a hole and Threads.	User Steam Sterilization
105		NT-0350		
106		NT-0400		
107		NT-0450		
108		NT-0500		
109		NT-0550		
110		NT-0600		
111		NT-0650		
112		NT-MS45		
113		NT-MS55		
114		NT-MS65		
115		NT-MS75		
116		CT-MS45		
117		CT-MS55		
118		CT-MS65		
119		CT-MS75		
120		NT-LK45		
121		NT-LK55		
122		NT-LK65		

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No.	Name	Model Number	Description	Note (Sterilization/Disposable)
123		CT-LK45		
124		CT-LK55		
125		CT-LK65		
126		NT-LK45P		
127		NT-LK55P		
128		NT-LK65P		
129	Guide Sleeve	GS-1395	It is temporarily mounted on Guide Bush when the surgical instrument or screw is guided and supported by the manipulator (User Steam Sterilization)	User Steam Sterilization
130	Screwdriver Marker	SM-0700	It is assembled with a compatible model of Screwdriver Body	User Steam Sterilization
131		SM-0800		
132		SM-0850		
133		SM-1000		
134		SM-1200		
135		SM-U001		
136		SM-0800S		
137		SM-1200S		
138	Screwdriver Body	LK-0001	It is used to insert a screw into a threaded hole	User Steam Sterilization
139		LK-0002		
140		LK-0003		
141	Handle	A1A-1047	It is assembled with Complatible model of handle use accessories	User Steam Sterilization
142		LS03-0304		

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No.	Name	Model Number	Description	Note (Sterilization/Disposable)
143	Instrument Container	CC-0000	It is used for storing the cleaning and sterilization surgical instruments	User Steam Sterilization
144		CM-0000		

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5.7. Complete description of the key functional elements, its formulation, its composition and its functionality

Please refer to the Section 5.2. Complete Description of the Medical Device

5.8. Explanation of any novel features

Not applicable

5.9. Labelled Pictorial Representation

Please refer to the Section 5.2. Complete Description of the Medical Device

5.10. Intended Use

CUVIS-spine is intended for use as an aid for precisely locating anatomical structures and for the spatial positioning and orientation of guide bush to be used by surgeons for navigating and/or guiding compatible surgical instruments in open or percutaneous pedicle screw placement provided that the required markers and rigid patient anatomy can be identified on 3D or 2D image.

5.11. Indications

CUVIS-spine is intended patient population is as following.

Age	Adult
Gender	No restrictions
Weight	No restrictions
Nationality	No restrictions
State	Having spine disease who requires spine surgery - There is no data is available to support CS200 safety and effectiveness when used at the C1-2 spinal levels
Health	Exclusion of patients under the following conditions - A patient with therapeutic / non-adjustable coagulation disorders - A patient who is pregnant or breastfeeding

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	- A patient who is judged to be unsuitable for the surgery at the discretion of medical doctor
--	--

5.12. Instruction for Use

Details may refer to **Appendix 1**.

- ✓ Appendix 1-1. User Manual (ENG) (Doc No. 310023)

5.13. Contraindications

- Use of this device is prohibited in cases where it is contravened in general surgical procedures
- Exclusion of patients under the following conditions: therapeutic / non-adjustable coagulation disorders, pregnant or breastfeeding, and judged to be unsuitable for the surgery at the discretion of the medical doctor.

5.14. Warnings and Precautions

1) Warnings

(1) Warning for electromagnetic compatibility

- Use of this equipment adjacent to or stacked with other equipment should be avoided because it could result in improper operation. If such use is necessary, this equipment and the other equipment should be observed to verify that they are operating normally.
- Use of accessories, transducers and cables other than those specified or provided by the manufacturer of this equipment could result in increased electromagnetic emissions or decreased electromagnetic immunity of this equipment and result in improper operation.
- Portable RF communications equipment (including peripherals such as antenna cables and external antennas) should be used no closer than 30cm (12inches) to any part of the CS200, including cables specified by CUREXO, INC. Otherwise, degradation of the performance of this equipment could result.

(2) Warning for cleaning and sterilization

- Do not use steel wool or abrasive cleaners as they may scratch metal finishes.

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- Do not use undiluted detergents or detergent solutions with pH greater than 8.0 as it can discolor the metal surface.
- Avoid solutions containing iodine or high chlorine content.
- Contaminated or used parts should not be cleaned with the parts loaded in an instrument container. They must be cleaned separately from the instrument container and tray.
- Long, narrow cannulations, blind holes and intricate parts require more particular attention during cleaning process.
- The sterilization parameters are only valid for devices that are adequately cleaned.
- Contaminated parts should be separated from non-contaminated parts to avoid contamination of personnel or surroundings.

2) Precautions

- This product must be used by qualified personnel to perform spinal surgery using a 3D or 2D imaging device, and the user must be trained by the manufacturer or the person authorized by the manufacturer.
- When using with 3D or 2D imaging device in step of scan-setup, move the manipulator only used by the function of hand-guide pressing or releasing foot switch to avoid collision.
- This product is intended to guide surgical instruments in spinal surgery. Do not connect it to the other parts that are not listed as accessory or compatible parts.
- Use of parts and devices that are not specified by the manufacturer may cause malfunction or damage to internal parts
- Avoid the use of other equipment that generates a high level of electromagnetic radiation in the vicinity of the equipment to prevent malfunction of the equipment due to potential electromagnetic or other interference. In unavoidable circumstances, contact the manufacturer or the person authorized by the manufacturer
- Use of equipment generating a high level of electromagnetic radiation may cause malfunction or damage to internal parts.
- If there is a noise or flashing of the touch screen monitor due to electrical fast transient (EFT), it is recommended that the user temporarily stops the operation of the touch screen monitor until the phenomenon is resolved.
- Operation of the touch monitor while a noise or flashing occurs in the monitor may

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cause malfunction or damage to internal parts.

- Generation of screw insertion path and completion of operation can be done by qualified person only.
- If excessive movement of the vertebra occurs during the surgical procedure, an error may occur between the planned path of screw and the position guided by the device. Therefore, it is necessary to take measures to minimize the vertebra's movement during the surgical procedure. Nevertheless, if a large movement of the target occurs, it is recommended to perform the procedure again from the scan step.
- The device should be connected to a power supply with protective ground.
- It is recommended to use the robotic arm only when the residual voltage of the internal battery is 24V or more. Otherwise, the manipulator cannot be manually moved and the lifting device cannot be operated in an emergency situation in which external power is cut off.
- To check the residual voltage of the battery, external power has to be disconnected. The residual voltage may not be displayed correctly when external power is supplied.
- When the battery is connected by user, it has a risk of malfunction or explosion. The battery can only be connected or replaced by the manufacturer or the person authorized by the manufacturer.
- When storing and transporting the robotic arm, the lift switch must be in the neutral position. Otherwise, the battery is completely discharged, which shortens the life of the battery or causes performance degradation.
- When the robotic arm is stored for a long time, the battery must be fully charged every 6 months by connecting to an external power source to maintain battery performance.
- Do not unplug the main power supply before turning off the power switch.
- Disassembly and replacement of fuses, power cords and other parts may only be carried out by the manufacturer or the person authorized by the manufacturer.
- Do not put the load exceeding 1kg in the drawer of the main console.
- Check if the ventilating fan is working normally after powering the robotic arm.
- The user has to check if the ventilating fan of the robotic arm is working normally

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prior to use. When it does not work or causes noise or vibration, stop using the robotic arm and contact the manufacturer or the person authorized by the manufacturer.

5.15. Potential Adverse Effects

As with conventional surgery, damage to the nerves and dura mater due to the invasion of the screw into the neural canal, resulting in neurological abnormalities (abnormal pain, decreased sensation, paralysis, urinary and fecal incontinence), cerebrospinal fluid leakage, bleeding due to vascular damage, and decreased blood pressure may occur.

5.16. Alternative Therapy

Conventional spine surgery procedures are available

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5.17. Materials

1) System

No.	Description		Standard or Specification
1	Power unit	Power Cords with Locking System	250VAC, 50/60Hz, 16A
2		Power Inlet IEC Appliance Inlet C20 with Filter	250VAC, 50/60Hz, 16A
3		Molded Case Circuit Breakers	220VAC, 50/60Hz, 30A
4		AC-DC SMPS	85-264VAC, 480W / 48V, 480W / 24V
5		PC SMPS	100-240VAC, 50/60 Hz
6		Lift Battery	24VDC, Lead Acid 1.2Ah
7	Control unit	Lift Control Box	100-240VAC, 50/60Hz
8		Motor Driver	50VDC, 5A
9			70VDC, 10A
10	Operating unit	Foot Switch	2 pedals
11	Software	SRC	Version: 2.0x
12		SPNe	Version: 1.0x
13	Main Console	OTS Camera	Polaris Vega
14		OTS Arm	Al, Acetal
15		Touch Monitor	27“, 12VDC, 28W
16		Main Cart	Mobile Cart
17		Drawer	Working load: 1kg
18		Configurable SMPS	85-264VAC
19	Staff Console (Option)	Touch Monitor	27“, 24VDC, 28W
20		Staff Cart	Mobile Cart

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2) Accessories

(1) Non-contact human body

No.	Description		Standard or Specification
1	Guide Bush	GB-0950	SUS630
		GB-1250	SUS630
		GB-1300	SUS630
		GB-1350	SUS630
		GB-1400	SUS630
		GB-1450	SUS630
		GB-1500	SUS630
		GB-1550	SUS630
		GB-1600	SUS630
		GB-0953	SUS630
		GB-1199	SUS630
		GB-1303	SUS630
		GB-1356	SUS630
		GB-1373	SUS630
		GB-1410	SUS630
		GB-1388	SUS630
		GB-1398	SUS630
		GB-1510	SUS630
		GB-1613	SUS630
2	Patient Marker	PM-F001	Al
		PM-F002	Al
3	Detector Calibrator	DC-GE001	Acetal, Al
		DC-GM001	Acetal, Al
		DC-SM001	Acetal, Al
		DC-DK001	Acetal, Al
		DLDC-U001	Acetal, Al
4	Source Calibrator	SC-001	Acetal, SUS304
5	Registration Tool	RT-001	Acetal
		RT-002	Acetal
6	Registration Tool Adapter	RA-001	SUS630
7	CUVIS-spine	Robotic Arm Drape	PE

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No.	Description		Standard or Specification
	Drape	Tool Drape	PE
		Detector Drape	PE
8	Marker Ball	MB-001	Ø11.5mm
		MB-002	Ø11.8mm
9	Marker Driver	MD-0000	Al, SUS630
10	Tapper Marker	TM-0950	Al, SUS630
		TM-1350	Al, SUS630
		TM-0950S	Al, SUS630
11	Screwdriver Marker	SM-0700	Al, SUS630
		SM-0800	Al, SUS630
		SM-0850	Al, SUS630
		SM-1000	Al, SUS630
		SM-1200	Al, SUS630
		SM-U001	Al, SUS630
		SM-0800S	Al, SUS630
		SM-1200S	Al, SUS630
12	Bur Marker	BM-0800	Al, SUS630
		CM-1325	Al, SUS630
		BAM	Al, SUS630
13	Lenke Straight Probe Marker	LSPM-0800	Al, SUS630
14	Sharp Awl Marker	SAM-0800	Al, SUS630
15	Instrument Container	CC-0000	Al
		CM-0000	Al
16	Adapter	PM-A001	SUS630
17	Pin Vise	PV-0001	SUS630
18	Slide Hammer	SH-0001	SUS630
19	Handle	A1A-1047	-
		LS03-0304	-

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(2) Contact human body

No.	Material Name		Material Name	Standard	Composition	Contact area
1	Clamp	PM-C001	Stainless Steel	ASTM A564	100%	Bone
		PM-C002	Stainless Steel	ASTM A564	100%	Bone
		PM-C011	Stainless Steel	ASTM A564	100%	Bone
		PM-C012	Stainless Steel	ASTM A564	100%	Bone
		PM-C013	Stainless Steel	ASTM A564	100%	Bone
2	Pin	PM-P100	Stainless Steel	ASTM A564	100%	Bone
		PM-P150	Stainless Steel	ASTM A564	100%	Bone
		PM-P200	Stainless Steel	ASTM A564	100%	Bone
3	Probe	PB-0000	Stainless Steel	ASTM A564	100%	Bone
4	Dilator	DL-1300	Stainless Steel	ASTM A564	100%	Bone
		DL-1775	Stainless Steel	ASTM A564	100%	Bone
		DL-1300-MSZ	Stainless Steel	ASTM A564	100%	Bone
		DL-1775-MSZ	Stainless Steel	ASTM A564	100%	Bone
		DL-1300-LKP	Stainless Steel	ASTM A564	100%	Bone
		DL-1775-LKP	Stainless Steel	ASTM A564	100%	Bone
		DL-1300-MSI	Stainless Steel	ASTM A564	100%	Bone
5	Serration -tip Dilator	SD-0500	Stainless Steel	ASTM A564	100%	Bone
		SD-2030	Stainless Steel	ASTM A564	100%	Bone
		SD-1630BA	Stainless Steel	ASTM A564	100%	Bone
6	Stylet Tapper Body	ST-0550	Stainless Steel	ASTM A564	100%	Bone
		ST-0600	Stainless Steel	ASTM A564	100%	Bone
		ST-0650	Stainless Steel	ASTM A564	100%	Bone
7	Tapper Body	NT-0300	Stainless Steel	ASTM A564	100%	Bone
		NT-0350	Stainless Steel	ASTM A564	100%	Bone
		NT-0400	Stainless Steel	ASTM A564	100%	Bone
		NT-0450	Stainless Steel	ASTM A564	100%	Bone
		NT-0500	Stainless Steel	ASTM A564	100%	Bone
		NT-0550	Stainless Steel	ASTM A564	100%	Bone
		NT-0600	Stainless Steel	ASTM A564	100%	Bone
		NT-0650	Stainless Steel	ASTM A564	100%	Bone
		NT-MS45	Stainless Steel	ASTM A564	100%	Bone
		NT-MS55	Stainless Steel	ASTM A564	100%	Bone
		NT-MS65	Stainless Steel	ASTM A564	100%	Bone
		NT-MS75	Stainless Steel	ASTM A564	100%	Bone
		CT-MS45	Stainless Steel	ASTM A564	100%	Bone
		CT-MS55	Stainless Steel	ASTM A564	100%	Bone

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No.	Material Name		Material Name	Standard	Composition	Contact area
		CT-MS65	Stainless Steel	ASTM A564	100%	Bone
		CT-MS75	Stainless Steel	ASTM A564	100%	Bone
		NT-LK45	Stainless Steel	ASTM A564	100%	Bone
		NT-LK55	Stainless Steel	ASTM A564	100%	Bone
		NT-LK65	Stainless Steel	ASTM A564	100%	Bone
		CT-LK45	Stainless Steel	ASTM A564	100%	Bone
		CT-LK55	Stainless Steel	ASTM A564	100%	Bone
		CT-LK65	Stainless Steel	ASTM A564	100%	Bone
		NT-LK45P	Stainless Steel	ASTM A564	100%	Bone
		NT-LK55P	Stainless Steel	ASTM A564	100%	Bone
		NT-LK65P	Stainless Steel	ASTM A564	100%	Bone
8	Screwdriver Body	MQ-0000	Stainless Steel	ASTM A564	100%	Bone
		UI-0000	Stainless Steel	ASTM A564	100%	Bone
		LK-0001	Stainless Steel	ASTM A564	100%	Bone
		LK-0002	Stainless Steel	ASTM A564	100%	Bone
		LK-0003	Stainless Steel	ASTM A564	100%	Bone
9	Bur	MR223-30BE	Stainless Steel	ASTM A276	100%	Bone
		AN223-30BE	Stainless Steel	ASTM A276	100%	Bone
		NS223-30BE	Stainless Steel	ASTM A276	100%	Bone
		ST223-30BE	Stainless Steel	ASTM A276	100%	Bone
		LI223-30BE	Stainless Steel	ASTM A276	100%	Bone
		AO223-30BE	Stainless Steel	ASTM A276	100%	Bone
		MR223-40BE	Stainless Steel	ASTM A276	100%	Bone
		AN223-40BE	Stainless Steel	ASTM A276	100%	Bone
		NS223-40BE	Stainless Steel	ASTM A276	100%	Bone
		ST223-40BE	Stainless Steel	ASTM A276	100%	Bone
		LI223-40BE	Stainless Steel	ASTM A276	100%	Bone
		AO223-40BE	Stainless Steel	ASTM A276	100%	Bone
		MR277-30BE	Stainless Steel	ASTM A276	100%	Bone
		MR277-40BE	Stainless Steel	ASTM A276	100%	Bone
		AOL300-30BE	Stainless Steel	ASTM A276	100%	Bone
		AOL300-40BE	Stainless Steel	ASTM A276	100%	Bone
		M8238-30BE	Stainless Steel	ASTM A276	100%	Bone
		M8238-40BE	Stainless Steel	ASTM A276	100%	Bone
		M8277-30BE	Stainless Steel	ASTM A276	100%	Bone
		M8277-40BE	Stainless Steel	ASTM A276	100%	Bone
		BA232-30BE	Stainless Steel	ASTM A276	100%	Bone

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No.	Material Name		Material Name	Standard	Composition	Contact area
		BA232-40BE	Stainless Steel	ASTM A276	100%	Bone
		AO277-30BE	Stainless Steel	ASTM A276	100%	Bone
		AO277-40BE	Stainless Steel	ASTM A276	100%	Bone
		NS255-30BE	Stainless Steel	ASTM A276	100%	Bone
		NS255-40BE	Stainless Steel	ASTM A276	100%	Bone
		AC232-30BE	Stainless Steel	ASTM A276	100%	Bone
		AC232-40BE	Stainless Steel	ASTM A276	100%	Bone
10	Lenke Straight Probe Body	PB-0001	Stainless Steel	ASTM A564	100%	Bone
11	Sharp Awl Body	AW-0000	Stainless Steel	ASTM A276	100%	Bone

5.18. Other Relevant Specifications and Descriptive Information

There is no other relevant specifications and descriptive information.

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6. SUMMARY OF DESIGN VERIFICATION AND VALIDATION DOCUMENTS

6.1. Pre-Clinical Studies

1) Biocompatibility Test

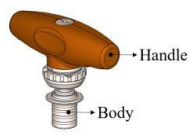
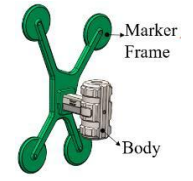
Among the system (Robotic Arm, Main Console, and Staff Console) and accessories that comprise CUVIS-spine, the parts that contact the human body are as follow.

No.	Part Description	Contact	Remark
1	Robotic Arm	No contact	System
2	Main Console	No contact	
3	Staff Console	No contact	
4	Guide bush	No contact	Accessory
5	Source Calibrator	No contact	
6	Registration Tool	No contact	
7	Registration Tool Adapter	No contact	
8	CS200 Drape	No contact	
9	Marker Ball	No contact	
10	Patient Marker	No contact	
11	Clamp	Bone	
12	Adapter	No contact	
13	Pin	Bone	
14	Pine Vise	No contact	
15	Slide Hammer	No contact	
16	Marker Driver	No contact	
17	Detector Calibrator	No contact	
18	Dilator	Bone	
19	Serration-tip Dilator	Bone	
20	Bur	Bone	
21	Probe	Bone	
22	Tapper	Bone	
23	Stylet Tapper	Bone	
24	Screwdriver	Bone	
25	Instrument Container	No contact	
26	Awl	Bone	
27	Lenke Probe	Bone	

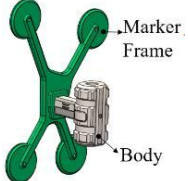
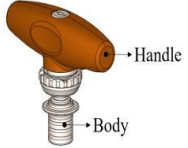
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No.	Part Description	Contact	Remark
28	Handle	No contact	

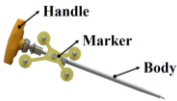
The table below shows the raw materials and contact types of components (accessories) that contact the patient's body among all the components listed in the table above.

Description	Details	Raw Material	Additives	Body Contact	Potential indirect patient contact	Remark
Clamp	-	Type 630 Stainless Steel	N/A	Bone	YES	
Pin	-	Type 630 Stainless Steel	N/A	Bone	YES	
Tapper	T-ratchet Handle - Body	Type 630 Stainless Steel	N/A	N/A	N/A*	
	T-ratchet Handle - Handle	Silicon	RBB-2881-70	N/A	N/A*	
	Tapper Marker - Body	Type 630 Stainless Steel	N/A	N/A	N/A*	
	Tapper Marker - Frame	Aluminum 6061	Green GM (Anodizing)	N/A	N/A*	
	Tapper Body	Type 630 Stainless Steel	N/A	Bone	YES	
Stylet Tapper	T-ratchet Handle - Body	Type 630 Stainless Steel	N/A	N/A	N/A*	

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Description	Details	Raw Material	Additives	Body Contact	Potential indirect patient contact	Remark
	T-ratchet Handle - Handle	Silicon	RBB-2881-70	N/A	N/A*	
	Tapper Marker - Body	Type 630 Stainless Steel	N/A	N/A	N/A*	
	Tapper Marker - Frame	Aluminum 6061	Green GM (Anodizing)	N/A	N/A*	
	Styler Tapper Body	Type 630 Stainless Steel	N/A	Bone	YES	
Screwdriver	T-ratchet Handle - Body	Type 630 Stainless Steel	N/A	N/A	N/A*	
	T-ratchet Handle - Handle	Silicon	RBB-2881-70	N/A	N/A*	
	Screwdriver Marker - Body	Type 630 Stainless Steel	N/A	N/A	N/A*	
	Screwdriver Marker-Frame	Aluminum 6061	TAC RED SCH (Anodizing)	N/A	N/A*	
	Screwdriver Body - Handle	Aluminum 6061	Sulfuric acid 98% (Anodizing)	N/A	N/A*	
	Screwdriver Body - Body	Type 630 Stainless Steel	N/A	Bone	YES	
Serration-tip Dilator	-	Type 630 Stainless Steel	N/A	Bone	YES	
Bur	-	TrimRite	N/A	Bone	YES	

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Description	Details	Raw Material	Additives	Body Contact	Potential indirect patient contact	Remark
	Bur marker	Aluminum 6061	Sulfuric acid 98% (Anodizing)	N/A	N/A*	
Probe	Probe-marker	Aluminum 6061	Sulfuric acid 98% (Anodizing)	N/A	N/A*	
	Probe-body	Type 630 Stainless Steel	N/A	Bone	YES	
Dilator	Shaft	Type 630 Stainless Steel	N/A	Bone	YES	
	Cannula	Type 630 Stainless Steel	N/A	Bone	YES	
Awl	T-ratchet Handle - Body	Type 630 Stainless Steel	N/A	N/A	N/A*	
	T-ratchet Handle - Handle	Silicon	RBB-2881-70	N/A	N/A*	
	Awl-Sharp awl marker	Aluminum 6061	TAC ORANGE LH	N/A	N/A*	
	Awl-Sharp awl body	TrimRite	N/A	Bone	YES	
Lenke probe	T-ratchet Handle - Body	Type 630 Stainless Steel	N/A	N/A	N/A*	
	T-ratchet Handle - Handle	Silicon	RBB-2881-70	N/A	N/A*	
	Lenke probe-straight probe marker	Aluminum 6061	SANODYE BLUE G (Anodizing)	N/A	N/A*	

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Description	Details	Raw Material	Additives	Body Contact	Potential indirect patient contact	Remark
	Lenke probe-straight probe body	Type 630 Stainless Steel	N/A	Bone	YES	

*Tapper / Stylet tapper / Screwdriver/ Probe/Bur Body/Awl/Lenke probe

According to the principle/method of operating these accessories, there is no possibility of the potential indirect patient contact except for the body. Because these accessories will be installed/inserted in the end-effector of the robotic arm.

Among the body contact accessories, there are two main raw materials for surgical tools that contact the human body as shown in the table above. One is SUS630 and other is TrimRite. Biological test results corresponding to the raw material of SUS630 and TrimRite.

The nature of body contact of these surgical tools is ‘implant device/bone’ with the contact duration less than 24 hours (A-limited (<24h)) as specified in ISO 10993-1. For these surgical tools, biocompatibility evaluation is performed in accordance with:

- ISO10993-1:2018 Biological Evaluation of medical devices – Part 1: Evaluation and testing within a risk management process
- ISO10993-5: 2009 Biological evaluation of medical device – Part 5: Tests for in vitro cytotoxicity
- ISO10993-10:2010 Biological evaluation of medical device – Part 10: Tests for irritation and skin sensitization
- ISO10993-11:2009 Biological evaluation of medical device – Part 11: Tests for systemic toxicity

We conducted Cytotoxicity, Sensitization, Intracutaneous Reactivity, Acute Systemic Toxicity, and Pyrogenicity test for each raw material. Details regarding each test reports are as below. Refer the attached file **Appendix 4**.

- ✓ Appendix 4-1. Biocompatibility Evaluation Report (Doc No. 300059)

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▪ **Raw material: SUS630 (Stainless Steel, Standard: ASTM A564)**

- Tasted Sample: Patient Marker (currently, renamed to ‘Clamp’)

As shown in 1.2 Contact Duration, the contact duration of the clamp is relatively high compared to the other parts. It is also produced under the same manufacturing process using the same raw material. So, the patient marker was used as a representative accessory for the test.

- Test Results

No.	Test Title	Report No.	Testing laboratory
1	Cytotoxicity	19-VL-0507	Chemon Inc.*
2	Sensitization	19-GI-0508	
3	Intracutaneous Reactivity	19-BL-0509	
4	Acute Systemic Toxicity	19-MA-0510	
5	Material mediated pyrogenicity	21-021679-01-1	KTL* (Korea Testing Laboratory)

▪ **Raw material: TrimRite (Stainless Steel, Standard: ASTM A276)**

- Tasted Sample: Schanz Screw**

- Test Results

No.	Test	Report No.	Testing laboratory
1	Cytotoxicity	19-068276-02-1	KTL* (Korea Testing Laboratory)
2	Sensitization		
3	Intracutaneous Reactivity		
4	Systemic Toxicity		
5	Material mediated pyrogenicity	20-080537-02-1	

* Each testing laboratories for biocompatibility test have been accredited in accordance with GLP (Good Laboratory Practice) standard, general requirements for the competence of testing laboratories.

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**** Comparison Table**

	Subjects Product	Equivalent Product
Product Name	Bur	Schanz screw
Picture		
Raw Material	TrimRite	
Contact	Implant Device / Bone	
Contact Duration	limited (≤ 24 h)	
Sterilization Condition	Steam / Pre-vacuum type / 132°C / 4 minutes	
Related system	CUVIS-spine (CUREXO, INC.)	CUVIS-joint (CUREXO, INC.)

- CUVIS-joint' is the orthopaedic surgical robot system manufactured by CUREXO, INC. same as 'CUVIS-spine'. Among the accessories used with 'CUVIS-joint', the Schanz screw is used in contact with the patient's bone such as femur or tibia during the surgery.

2) Physical Test Data

CUVIS-spine (CS200) has been evaluated through several methods; especially the following verification and validation tests were conducted to ensure the performance of the implemented features;

No.	Performance test	Description
1	310053 (Rev 0) Performance Test Report - Pose Repeatability -	Verifying that CS200 satisfies the performance requirement in terms of pose repeatability
2	310054 (Rev 0) Performance Test Report - Pose Accuracy -	verifying that CS200 satisfies the performance requirement in terms of pose accuracy.
3	310055 (Rev 0) Performance Test Report - Workspace -	verify that CS200 satisfies the performance requirement in terms of workspace..

Please refer to attached file **Appendix 5**.

- ✓ Appendix 5-1. Performance Test Report - Pose Repeatability (Doc No.310053)
- ✓ Appendix 5-2. Performance Test Report - Pose Accuracy (Doc No.310054)
- ✓ Appendix 5-3. Performance Test Report - Workspace (Doc No.310055)

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3) Compliance tests

The basic electrical safety and electromagnetic compatibility (EMC) required for this medical device have been verified as in the following reports. All the specified test items are inspected to the applied international and national standards. Following is the report tested by certified laboratories.

Refer to the Appendix in the table below.

No.	Title	Standard	Report No.
1	EMC Test	- IEC 60601-1-2:2014/AMD1:2020 - IEC TS 60601-4-2:2024	310403 (Rev 3) EMC Test Report
2	Safety Test	- IEC 60601-1:2005/AMD2:2020 - IEC 60601-1-6:2010/AMD2:2020 - National standard AAMI ES60601-1:2005/AMD2:2021 - IEC 80601-2-77:2019 - IEC 62366-1:2015/AMD1:2020 - IEC 62304:2006/AMD1:2015 - ISO 14971:2019	310404 (Rev 3) Safety Test Report

- ✓ Appendix 6-1. EMC Test Report (Doc No.310403)
- ✓ Appendix 6-2. Safety Test Report (Doc No.310404)

6.2. Software Validation Studies

The unit test, the integration test, and the system test were performed in each software, and validation report to verify the results and validity of each test can be checked as below table. CUVIS-spine software tests were conducted to satisfy the requirements of the IEC 62304.

Refer to the attached file Appendix 7.

- ✓ Appendix 7-1. Software Verification Validation Report (Doc No.310027)

6.3. Devices Containing Biological Material

Not Applicable

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6.4. Sterilization Validation Reports

1) Steam sterilization

Some accessories used with CUVIS-spine should be autoclaved before use by user.

No.	Part/Accessories	Remark
1	End-effector(male)	Part of CUVIS-spine system
2	Guide Bush	Accessories
3	Registration Tool Adapter	Accessories
4	Patient Marker	Accessories
5	Clamp	Accessories
6	Adapter	Accessories
7	Pin	Accessories
8	Pin Vise	Accessories
9	Slide Hammer	Accessories
10	Marker Driver	Accessories
11	Dilator	Accessories
12	Serration-tip Dilator	Accessories
13	Bur**	Accessories
14	Awl*	Accessories
15	Probe	Accessories
16	Lenke Probe*	Accessories
17	Tapper*	Accessories
18	Stylet Tapper*	Accessories
19	Screwdriver*	Accessories
20	Instrument Container	Accessories

* It consists of three parts that are sterilized separately.

- Awl: T-Ratchet Handle + Awl Marker + Awl Body
- Lenke Probe: T-Ratchet Handle + Lenke Probe Marker + Lenke Probe Body
- Tapper: T-Ratchet Handle + Tapper Marker + Tapper Body
- Stylet Tapper: T-Ratchet Handle + Tapper Marker + Stylet Tapper Body
- Screwdriver: T-Ratchet Handle + Screwdriver Marker + Screwdriver Tapper Body

** In the case of a bur (navigation type) that can be used with a bur marker, sterilize the bur and bur marker separately.

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It is sterilized using steam sterilization method as follows.

No.	Item	Description	
1	Sterilization Method	Pre-vacuum type	
2	Sterilization Cycle	Temperature	132°C
		Time	4 minutes
		Dry Time	20 minutes
3	Sterility Assurance Level (SAL)	10 ⁻⁶	

The sterilization validation confirmed that the sterilization cycle maintained a sterility assurance level (SAL) of 10⁻⁶ in accordance with the following standard.

- ANSI/AAMI ST79:20/A4:2013, Comprehensive guide to steam sterilization and sterility assurance in health care facilities, Amendment 3
- ISO 17665-1:2006, Sterilization of healthcare products–Moist heat–Part 1: Requirements for the development, validation and routine control of a sterilization process for medical devices

Refer to the attached file **Appendix 8**.

- ✓ Appendix 8-1. User Cleaning & Sterilization Evaluation Report (Doc No.300051)

2) EO (Ethylene Oxide) Sterilization

Among accessories for CUVIS-spine, the Robotic Arm Drape, Tool Drape and Detector Drape are sterile device. Each drapes are the sterile drape that covers each parts (Robotic Arm, source calibrator or registration tool, Detector calibrator) in sterilized zone. And, Marker Ball (Model name: MB-001, MB-002) are also sterile devices. Marker Ball Marker Balls are purchased product, each manufacturer has been conducted EO Sterilization validation for Marker ball.

No.	Name	Model Name	Human Body Contact
1	CUVIS-spine Drape	Robotic Arm Drape	No
2		Tool Drape	No
3		Detector Drape	No
4	Marker Ball	MB-001	No
5		MB-002	No

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The EO (Ethylene Oxide) sterilization cycle provides a sterility assurance level of 10^{-6} in accordance with ISO 11135:2014, Sterilization of health-care products – Ethylene oxide – Requirements for the development, validation and routine control of a sterilization process for medical devices. Also, the acceptance criteria for EO residue medical devices sterilized by EO gas is in accordance with ISO 10993-7:2008, Biological evaluation of medical devices–Part 7: Ethylene oxide sterilization residuals.

The EO sterilization residuals test and results are summarized in Ethylene Oxide Sterilization Validation Report. Refer to the attached file **Appendix 9**.

- Appendix 9. Sterility Test

No.	Part/Accessories	Test Title	Report No.
1	Drape	Validation Report	300075 (Rev 1)
2	Marker Ball (MB-001)	Validation Protocol	MVP2019-027-PQ
3		Validation Report	MVP2019-027-PQ
4		Bioburden Methods Validation	VP20-IZI-BB-03
5		ETO Residual Analysis	171208-1 RES
6	Marker Ball (MB-002)	Validation Report	2IMK1-21V1
7		Validation Report	2IMK2-21V1
8		Validation Report	iiMED R0010

- ✓ Appendix 9-1. Drape EO Sterilization Validation Report (Doc No.300075)
- ✓ Appendix 9-2. Marker Ball EO Sterilization Validation Protocol (Doc No.MVP2019-027-PQ)
- ✓ Appendix 9-3. Marker Ball EO Sterilization Validation Report (Doc No.MVP2019-027-PQ)
- ✓ Appendix 9-4. Marker Ball Bioburden Methods Validation (Doc No.VP20-IZI-BB-03)
- ✓ Appendix 9-5. Marker Ball ETO Residual Analysis (Doc No.171208-1 RES)
- ✓ Appendix 9-6. Marker Ball EO Sterilization Validation Report (Doc No.2IMK1-21V1)
- ✓ Appendix 9-7. Marker Ball EO Sterilization Validation Report (Doc No.2IMK2-21V1)
- ✓ Appendix 9-8. Marker Ball EO Sterilization Validation Report (Doc No.iiMED R0010)

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6.5. Determination of Shelf Life

Regarding the products that are sterilized and provided, stability tests were conducted to determine the shelf life.

The stability test is performed on two sterilized and supplied single-use accessories, Drape and Marker Ball. Shelf-life is normally considered to be the time the device can be kept in the packaging prior to its first use, and the shelf-life testing is based on accelerated age testing and real time testing. Refer to the attached files **Appendix 10**.

- Appendix 10. Stability Test

No.	Part/Accessories	Test Title	Report No.
1	Drape	Shelf Life Test (Accelerated aging 3 years) Report	CRX-SLR-2303
2		Shelf Life Test (Real time 3 years) Report	CRX-SLP-2304
3	Marker Ball (MB-001)	Accelerated Aging Test Report (36month)	DV-DPA Accelerated Aging-2014 Rev 0
4		Process Verification Stability Study(Real time 41month)	PV-ETO Stability for Spherz-2018 Rev B
5	Marker Ball (MB-002)	Accelerated Aging Test Report (5 years)	-
6		Shelf Life Test Report (Real time 5 years)	0901091

According to the test results, the shelf life of each product is as following.

- Drape: 3 years
 - Marker Ball (MB-001): 3 Years
 - Marker Ball (MB-002): 5 Years
-
- ✓ Appendix 10-1 Shelf Life Test (Accelerated aging 3 years) Report (Doc No. CRX-SLR-2303)
 - ✓ Appendix 10-2. Shelf Life Test (Real time 3 years) Report (Doc No. CRX-SLP-2304)
 - ✓ Appendix 10-3. Accelerated Aging Test Report (36month) (Doc No. DV-DPA Accelerated Aging-2014 Rev 0)
 - ✓ Appendix 10-4. Process Verification Stability Study (Doc No. PV-ETO Stability for Spherz-2018 Rev B)
 - ✓ Appendix 10-5. Accelerated Aging Test Report (5 years)
 - ✓ Appendix 10-6. Shelf Life Test Report (Real time 5 years) (Doc No. 0901091)

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7. CLINICAL EVIDENCE

7.1. Use of Existing Bibliography (Literature Review)

Refer to the attached files **Appendix 11**.

- ✓ Appendix 11-1 Clinical Evaluation Report (Doc No. 300024)

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8. MEDICAL DEVICE LABELLING

8.1. Samples of Labels on the Device and its Packaging

Regarding the detailed description, refer to the attached file **Appendix 1**.

1) System Label

- Applied to: Robotic Arm, Main Console, Staff Console
- Detailed information: '*' or  These markings mean that the contents are subject to change for each product.



No.	Item	Detail
1	Model Name	CS200
2	Manufacture Date*	YYYY.MM.DD
3	Manufacturer	CUREXO, INC.
4	Serial No.*	CS02XXX 'XXX' : Manufacturing sequence number
5	Safety Symbol	- B type Applied part - Refer to Manual - Waste sign
6	Barcode *	GS1-128 or GS1 Data Matrix Type

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2) Main Component Label

- Applied to: Robotic Arm, Main Console, Staff Console
- Detailed information: '*' or  These markings mean that the contents are subject to change for each product.

Robotic Arm
REF CB-10000-000 **SN** CS02100001
Weight 266kg  2022-01-07
Rating 100-240VAC, 50/60Hz, 1,500VA

(01)08800043830000(11)220107(21)CS02001

Main Console
REF CB-20000-000 **SN** CS02200001
Weight 80kg  2022-01-07
Rating 100-240VAC, 50/60Hz, 280VA

(01)08800043830000(11)220107(21)CS02001

Staff Console
REF CB-30000-000 **SN** CS02300001
Weight 35kg  2022-01-07
Rating 100-240VAC, 50/60Hz, 65VA

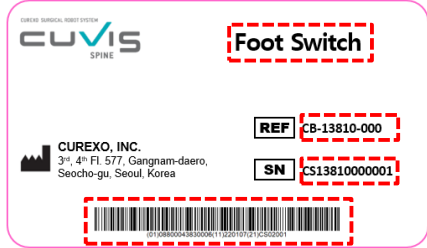
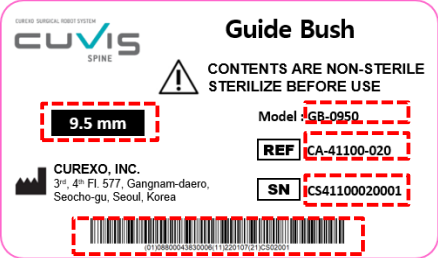
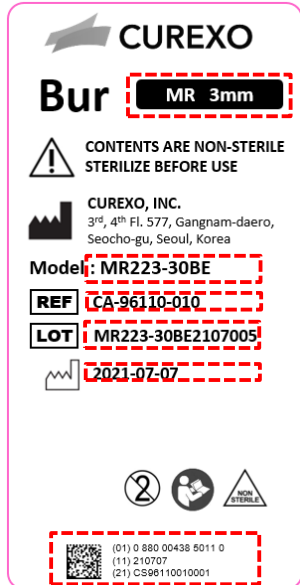
(01)08800043830000(11)220107(21)CS02001

No.	Item	Detail
1	Model Name	Robotic Arm, Main Console, Staff Console
2	Reference Number (REF)	Part Number
3	Manufacture Date)*	YYYY-MM-DD
4	Manufacturer	CUREXO INC.
5	Serial No.*	Manufacturing sequence number
6	Rating	Electric Voltage, Volt-Ampere
7	Weight	
8	Barcode*	GS1-128 or GS1 Data Matrix Type

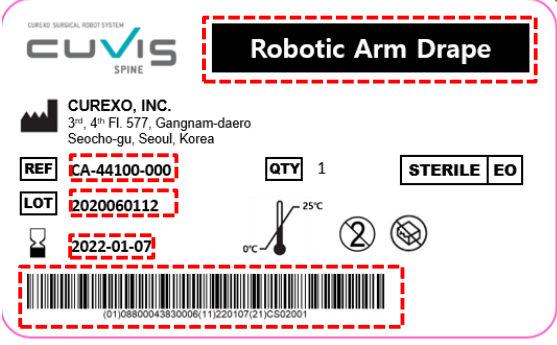
3) Accessory Packaging Label

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'*' or These markings mean that the contents are subject to change for each product.

No.	Group	Label Example	Remark
1	System Component	 The label for the Foot Switch includes the CUREXO logo, the text 'Foot Switch', a reference number (REF) CB-13810-000, a serial number (SN) CS13810000001, and a barcode.	Used for components such as Foot Switch, etc.
2	Accessories-Reusable user sterilization products	 The label for the Guide Bush includes the CUREXO logo, the text 'Guide Bush', a warning symbol and 'CONTENTS ARE NON-STERILE STERILIZE BEFORE USE', a size of 9.5 mm, a model number (MR-0950), a reference number (REF) CA-41100-020, a serial number (SN) CS41100020001, and a barcode.	
3	Accessories-Disposable user sterilization products	 The label for the Bur includes the CUREXO logo, the text 'Bur', a size of MR 3mm, a warning symbol and 'CONTENTS ARE NON-STERILE STERILIZE BEFORE USE', the CUREXO, INC. address, a model number (MR223-30BE), a reference number (REF) CA-96110-010, a lot number (LOT) MR223-30BE2107005, a date (2021-07-07), and a barcode.	Bur

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No.	Group	Label Example	Remark
4	Surgical Tools – Disposable sterilized products		Drape

4) Attachment location

(1) System Label and Main Component Label

- ① Robotic Arm: Right area above the port panel of the robotic arm
- ② Main Console: Lower area of rear right surface of the main console
- ③ Staff Console: Lower area of rear right surface of the staff console



[Robotic Arm]



[Main Console]



[Staff Console]

(2) Accessories

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Surface of the packaging



8.2. Instructions for use, training materials & instructions for installation and maintenance

Refer to the attached files **Appendix 1**.

- ✓ Appendix 1-1. User Manual (ENG) (Doc No. 310023)
- ✓ Appendix 1-2. System Labeling & Packaging Specification (Doc No.310031)
- ✓ Appendix 1-3. Service Manual (ENG) (Doc No.310080)
- ✓ Appendix 1-4. CUVIS-spine Catalog (ENG) (Doc No.900033)

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9. RISK ANALYSIS

Risk analysis was done and this report describes the result of risk management process for CUVIS-spine in accordance with ISO 14971:2019.

Risk analysis was conducted based on the risk management plan, which documented the results of the risk management process for the development phase.

The risk analysis was performed and documented using the DFMEA Table and PFMEA Table. Refer to the attached files **Appendix 12**.

- ✓ Appendix 12-1. Risk Management Plan (Doc No. 310003)
- ✓ Appendix 12-2. Risk Management Report (Doc No. 310005)

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10. MANUFACTURER INFORMATION

10.1. Manufacturing Process

Refer to the attached file **Appendix 13.**

- ✓ Appendix 13-1. Manufacturing Process (Doc No. 310028)

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11. Certificate

Refer to attached files **Appendix 14.**

- ✓ Appendix 14-1. CUREXO ISO13485 Certificate

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12. Appendix

Appendix No.	Document Name
Appendix 1	1-1. User Manual (ENG) (Doc No. 310023)
	1-2. System Labeling & Packaging Specification (Doc No.310031)
	1-3. Service Manual (ENG) (Doc No.310080)
	1-4. CUVIS-spine Catalog (ENG) (Doc No.900033)
Appendix 2	2-1. Certificate of Republic of Korea
	2-2. Certificate of US
	2-3. Certificate of Indonesia
	2-4. Certificate of Taiwan
	2-5. Certificate of Malaysia
Appendix 3	3-1. Declaration of AE and FSCA
Appendix 4	4-1. Biocompatibility Evaluation Report (Doc No. 300059)
Appendix 5	5-1 Performance Test Report - Pose Repeatability (Doc No.310053)
	5-2. Performance Test Report - Pose Accuracy (Doc No.310054)
	5-3. Performance Test Report - Workspace (Doc No.310055)
Appendix 6	6-1. EMC Test Report (Doc No.310403)
	6-2. Safety Test Report (Doc No.310404)
Appendix 7	7-1. Software Verification Validation Report (Doc No.310027)
Appendix 8	8-1. User Cleaning & Sterilization Evaluation Report (Doc No.300051)
Appendix 9	9-1. Drape EO Sterilization Validation Report (Doc No.300075)
	9-2. Marker Ball EO Sterilization Validation Protocol (Doc No.MVP2019-027-PQ)
	9-3. Marker Ball EO Sterilization Validation Report (Doc No.MVP2019-027-PQ)
	9-4. Marker Ball Bioburden Methods Validation (Doc No.VP20-IZI-BB-03)
	9-5. Marker Ball ETO Residual Analysis (Doc No.171208-1 RES)
	9-6. Marker Ball EO Sterilization Validation Report (Doc No.2IMK1-21V1)
	9-7. Marker Ball EO Sterilization Validation Report (Doc No.2IMK2-21V1)
	9-8. Marker Ball EO Sterilization Validation Report (Doc No.iiMED R0010)
Appendix 10	10-1 Shelf Life Test (Accelerated aging 3 years) Report (Doc No. CRX-SLR-2303)
	10-2. Shelf Life Test (Real time 3 years) Report (Doc No. CRX-SLP-2304)
	10-3. Accelerated Aging Test Report (36month) (Doc No. DV-DPA Accelerated Aging-2014 Rev 0)

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	10-4. Process Verification Stability Study (Doc No. PV-ETO Stability for Spherz-2018 Rev B)
	10-5. Accelerated Aging Test Report (5 years)
	10-6. Shelf Life Test Report (Real time 5 years) (Doc No. 0901091)
Appendix 11	11-1 Clinical Evaluation Report (Doc No. 300024)
Appendix 12	12-1. Risk Management Plan (Doc No. 310003)
	12-2. Risk Management Report (Doc No. 310005)
Appendix 13	13-1. Manufacturing Process (Doc No. 310028)
Appendix 14	14-1. CUREXO ISO13485 Certificate